

# GraphDecide: Benchmarking Decision Models on Graphs

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## Abstract

Graph applications often require repeated decisions, for which invoking generative large language models (LLMs) can incur substantial latency. System One models offer a promising route to fast action selection, yet their effectiveness on graph tasks remains unclear. We introduce GraphDecide, a benchmark for assessing whether decision-oriented models can reason about graph structure and construct useful complete solutions. The benchmark combines complete construction on public traveling salesperson and signed maximum-cut instances with controlled graph queries and synthetic construction tasks. On the primary public panel, Jev achieves smaller tour gaps than the tested bounded Qwen configurations, but this advantage does not extend to signed maximum cut. Classical heuristics produce substantially better solutions on both tasks. Controlled queries further show that reliable edge recognition does not ensure broader structural correctness. Together, these complementary evaluations connect individual decisions to complete outcomes and reveal task-dependent strengths and limitations, providing a basis for assessing fast-action models beyond response format or intended speed.

## 1 Introduction

Graph applications often require many successive decisions, so the latency of invoking generative large language models (LLMs) can accumulate across a complete solution. This motivates interest in “System One” models, which promise rapid decisions without generating long reasoning traces. Jev exemplifies this direction: its interface returns structured choices and associated probabilities, and its introduction emphasizes low response latency [1]. Open decision-oriented systems offer related bounded readouts [4, 9, 15, 25]. For graph algorithms, such interfaces suggest a simple building block: supply the current graph state and let a model choose the next action. Whether these choices support reliable graph computation, however, is not established by their format or intended speed.

Graph computation tests more than one capability. Recognizing an edge requires a local lookup; deciding whether a vertex disconnects a graph requires reasoning about alternative paths. Construction introduces a different demand: every selected city can be legal while the resulting traveling salesperson problem (TSP) tour is unnecessarily long. A useful evaluation must therefore connect individual decisions to complete outcomes. It must also retain unsuccessful episodes: excellent solutions on the cases that finish need not imply a reliable solver.

GraphDecide connects this capability question to two complementary tests (Figure 1). On published TSPLIB tour and OPTSICOM signed-cut instances, a model must turn repeated choices

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into a complete solution that can be compared with established references and classical algorithms. Controlled graph queries then distinguish direct lookups from structural reasoning; small synthetic constructions support exact analysis of the resulting decision histories. The first setting measures complete solution quality, while the second helps characterize what the model gets right or wrong about graphs.

We ask three research questions (RQs): **RQ1:** How good are complete constructed solutions, how often does a configuration finish, and what runtime is observed? **RQ2:** Which local and structural graph queries can it answer correctly? **RQ3:** Which findings persist on new graphs and under changes to presentation or interface? LLMs are included as bounded baselines and separately budgeted reasoning references, not assumed to share Jev’s design. Our contributions are:

1. **A graph-specific decision benchmark.** Six query tasks and four construction tasks connect structured actions to graph correctness and solution quality under explicit interaction contracts.
2. **Measurements of correctness and complete outcomes.** Five configurations complete 255 primary public-instance episodes across seventeen graphs and three paired conditions, evaluated against classical references. A separate twelve-configuration synthetic matrix diagnoses graph capability. Queries use exact accuracy; constructions use objective gap and completion coverage. Paired classical comparisons distinguish how often a method wins from how large its quality gap remains.
3. **Controlled interpretation of graph capability.** Matched query pairs, new-graph confirmation and presentation controls test robustness; probability audits separate action accuracy from probability quality. Exact continuation analysis provides a supplementary diagnostic on small graphs, rather than a claim about optimal planning at public-instance scales.

## 2 Related work

**Graph reasoning and optimization.** NLGraph evaluates natural-language graph reasoning; GraphArena broadens graph-computation evaluation to real-world-derived instances and verified solution quality [24, 28]. GraphWiz and GraphLLM study graph-specific training and representations [2, 3]. The Cormen–Leiserson–Rivest–Stein (CLRS) benchmark targets algorithmic learning [27]. Learned constructive graph policies and attention-based routing also predate the present decision interfaces [12, 13]. HeurAgenix develops and selects heuristics for combinatorial optimization [29]; GraphDecide instead evaluates direct choices of individual cities, vertex assignments or seeds, not which heuristic to run. GraphDecide studies general-purpose bounded choices, rather than proposing a graph-trained policy. Its common interaction separates a correct answer, a legal completion and a good objective value.

**Decision-oriented models and evaluations.** Jev’s typed interface and the SemIf, Decider, Kev and Laya systems motivate our comparison [1, 4, 9, 15, 25]. JevBench evaluates typed choices across application domains [5]; WorldTour already explores Jev-guided next-city selection [14]. JevAdvBench examines models described as using reinforcement learning for calibrated decisions (RLCD), measuring decision flips relative to repeated-call noise on 812 questions in 66 scenarios for one Jev version [8]. Our question concerns graph correctness and complete solution quality across configurations, not only sensitivity to perturbations. We do not assume that all evaluated systems use RLCD or equivalent architectures. Appendix I provides further context.

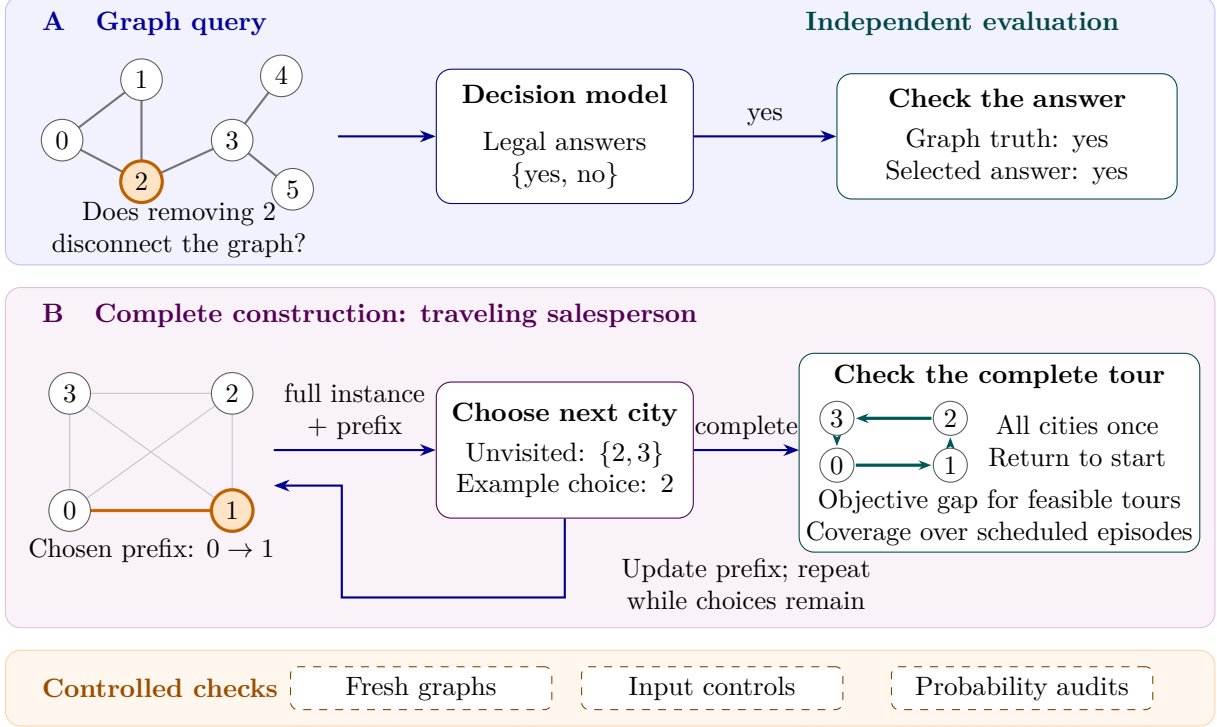


Figure 1: GraphDecide’s decision-and-evaluation loop: illustrative states, not benchmark episodes. A query is scored against graph truth. Primary construction repeatedly exposes the full instance and actual prefix with all legal next-city choices, ending with the forced last city and return. Independent evaluators report correctness or completed-solution gap plus scheduled coverage; no reference guides the model. The lower strip denotes separate controlled comparisons.

### 3 Tasks and evaluation design

Models receive graph structure, a question or current construction state, and legal action identifiers (IDs) and descriptions. Reference answers, generation labels and objective optima are withheld. A query ends after one choice; a construction updates the model-visible state with the model’s actual choice and continues. This shared interface permits comparison without substituting oracle prefixes for the histories that a model produces.

**Query tasks.** The six queries cover adjacency, degree, cycle existence, connectivity, thresholded shortest-path distance, and articulation. An articulation vertex is one whose deletion increases the number of connected components. Degree offers all  $n$  answers from 0 to  $n - 1$ ; the other tasks are binary. Graphs are simple, undirected and unweighted, including isolated vertices. Exact contracts appear in Appendix A.

**Construction tasks.** Maximum cut (MaxCut) assigns vertices to two sides to maximize crossing edge weight: unit weights in the synthetic bank, signed weights in the public panel. TSP constructs a closed tour by choosing the next unvisited city. Deterministic linear-threshold (LT) influence maximization selects two distinct seeds to maximize final activated nodes. Modularity construction assigns vertices to at most two communities to maximize resolution-1 modularity [18]. Synthetic cut and modularity fix vertex 0 in the first group and assign the remaining vertices in order; empty

groups are legal. Primary TSP runs start at the supplied city (0 in the synthetic bank), expose every unvisited city as a legal next choice, and force the final city and return. The synthetic TSP instances use Manhattan distance. Public TSP instances retain their specified distance convention, rather than inheriting that synthetic metric. For LT, incoming edge weights are  $1/d^-(v)$  and the fixed threshold is  $1/2$ ; both seeds activate simultaneously, followed by synchronous diffusion. This deterministic variant does not inherit the random-threshold model’s submodularity guarantee [11].

**Synthetic data and verification.** The diagnostic bank contains 960 cases: 800 exact queries and 160 constructions at sizes 8–12 (Table 1). Development contains 356 realized cases across sizes 5–12. Heterogeneous generators vary topology, and selected binary query pairs match degree-based attributes while having opposite labels. Distance pairs have finite distances exactly  $h$  and  $h + 1$ . Graph identity checks exclude overlap with a recorded historical pool. These controls target specified shortcuts and repeated structures, not all possible training-data contamination.

Ground truth and objectives are computed separately from model requests. Adjacency, distance and articulation oracles were tested on all 1,024 labeled five-vertex graphs; TSP uses exact subset dynamic programming, and partition and seed objectives permit exhaustive checks. A separate new-seed, topology-excluded confirmation realizes 952 of 960 requested cases, with eight cycle cases missing from generation. Only Jev, Qwen3.5-4B/9B grammar generation and Qwen3.5-4B candidate scoring receive this confirmation; B denotes billions of parameters. Full generation counts, chronology and independent within-project checks are in Appendix B.

## 4 Models, metrics and execution

**Configurations and budgets.** The synthetic matrix has ten direct-decision configurations: Jev’s native selection; no-thinking Qwen3.5-4B/9B/27B grammar generation [21]; Qwen3.5-4B candidate-token scoring; trained Decider-2B v11, Kev-4B and Laya; and Qwen3.8-27B [22] and Qwen2.5-72B-Instruct [20]. The last two use bfloat16 (BF16) weights. Grammar generation is greedy over legal choices with a 64-token ceiling. Qwen3.8 thinking is disabled; Qwen2.5 uses its standard instruct template. The larger/newer local configurations have a 180-second request deadline. GPT-5.4 and GPT-6-Astra are separate reasoning references, with tool-free default reasoning, a 4,096-token ceiling and a 180-second deadline. These settings are not matched-compute comparisons. Checkpoints, renderings and compatibility details are in Appendix C; the public-instance panel has its own recorded configuration roster.

**Failures and action semantics.** The same initial logical request is used across configurations in a panel. Subsequent states follow actual model choices, without retries, answer repair or revisions to earlier actions. Eligibility follows each experiment’s registered adapter policy. The public native-action rule in Section 5 does not retrospectively change the synthetic protocols, including earlier strict-response controls. Unsupported inputs, timeouts and invalid outputs remain in scheduled denominators. Feasibility means a valid query answer or a complete legal construction, not correctness or optimality.

**Correctness, quality and coverage.** Synthetic query accuracy is the equal-weight mean of five size-specific accuracies; degree’s pooled accuracy differs because each size has  $8n$  queries. Synthetic construction quality uses additive gap:  $f_i^* - f_i$  for maximization and  $f_i - f_i^*$  for minimization. Mean gap is conditional on complete feasible episodes and is always paired with feasible coverage over all scheduled episodes. All tied optima are accepted. Synthetic gaps have task-specific units and

are not percentages. The public panel reports relative gaps, separately by task:  $100(f_i - r_i)/r_i$  for TSP and  $100(r_i - f_i)/r_i$  for MaxCut. TSP’s  $r_i$  is a published proven optimum; MaxCut’s is a positive, source-reported historical best-known value. Cut gaps are never clipped: values above 100% correspond to negative achieved objectives, and negative gaps improve the historical reference. Each public mean first averages the three conditions within an instance, then weights instances equally; complete balanced coverage also makes it equal to the episode mean. Tasks are never pooled. Paired win/tie/loss counts compare achieved objectives on the same public instance and condition, with exact integer ties and the appropriate minimization or maximization direction. These frequencies complement, but do not replace, gap magnitudes. Synthetic optimum-hit counts are supplementary (Appendix N).

**Uncertainty.** The twelve-configuration comparison is descriptive, not a family-wise significance claim. Original four-configuration intervals use 2,000 size-stratified paired bootstrap replicates over clusters joining graph identity and matched-pair relations (Appendix C). Generation shortfalls have no prediction and are not model errors; execution failures on generated cases remain in the denominator.

## 5 Public-instance construction results

**Instances and direct-decision protocol.** Seven full TSPLIB instances contain 100–225 cities with nearest-integer Euclidean distances and published proven optima [23]. Every unvisited city remains a legal choice; there is no candidate filtering. Ten OPTSICOM Set2 spin-glass instances each have 125 vertices, 375 edges and weights  $-1$  or  $+1$  [17, 19]. Their references are source-reported historical best-known values from a pinned 2016 snapshot, not proven optima or current global records. Jev and four Qwen3.5 grammar configurations receive three paired relabeling/start conditions. Qwen uses no-thinking legal-choice generation with a 16-token ceiling; at most four episodes run concurrently per model. All GPT results in this paper belong to the synthetic experiments. The public Jev protocol scores its legal native action independently of probability-vector conformance, without normalization, retries or argmax replacement. The synthetic protocols are unchanged. Appendix F details the contracts and excluded calibration.

Classical references range from sampled uniform legal choices to nearest neighbor (NN) or incremental greedy construction, followed by two-edge exchange (2-opt) or single-vertex-flip improvement, respectively.

**RQ1: Complete construction does not imply useful quality.** All 255 primary scheduled model episodes finish: 21 tours and 30 cuts per configuration. Jev has a smaller TSP mean gap (66.01%) than all four bounded Qwen configurations (307.11–750.53%), but remains well behind nearest neighbor (27.65%) and its 2-opt refinement (5.93%). The signed-cut ordering is different: Jev’s 104.26% reference gap is larger than all four Qwen gaps (97.31–100.64%) and the sampled random reference (100.28%). Incremental greedy and single-flip refinement achieve 29.20% and 22.02%. Gaps above 100% are legitimate for signed cuts, not missing completions. The classical refinements can revise choices: they compare complete solution quality, not equal-budget online decision making.

The paired outcomes sharpen this contrast (Figure 2). Jev beats the sampled random tour in all 21 TSP conditions, but loses to nearest neighbor and to its 2-opt refinement in every condition. On signed cuts it wins nine, ties four and loses seventeen comparisons with sampled random, and loses all thirty comparisons with each greedy or local-search reference.

## Jev versus classical references

Paired final objectives on identical graphs and relabeling/start conditions

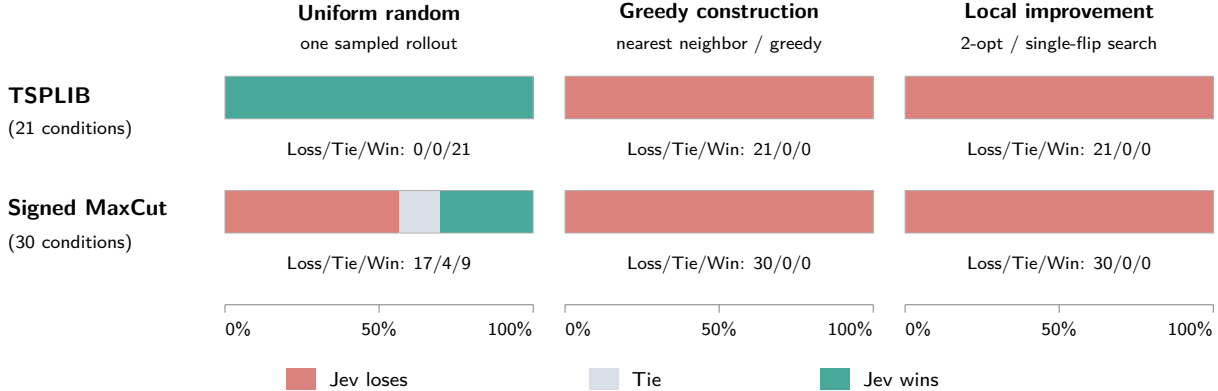


Figure 2: Primary full-action complete-solution comparisons: Jev’s native action policy versus three classical references. A shorter tour or a larger signed cut wins; ties use exact integer objectives. Bars contain 21 TSP or 30 MaxCut instance–condition pairs from seven and ten graphs, respectively. The three conditions per graph are correlated, not independent trials. Random is sampled, not an exact policy expectation. Win frequency and gap magnitude answer different questions; aggregate gaps appear in Table 3.

All 255 model and 153 classical objectives were recomputed independently. Instance-level values and provenance appear in Appendix G.

**Audited episode wall time.** Jev’s median audited episode wall times are 20.95 seconds for TSP and 21.25 seconds for MaxCut, versus 48.76–154.56 and 33.52–71.29 seconds across the Qwen configurations. These include request construction, model calls and response validation, ledger logging, terminal scoring, and post-run trajectory replay checks before sealing; they are not pure solver or inference times. Different hardware and cloud paths, with up to four concurrent episodes per model, make these comparisons descriptive, not intrinsic architectural speedups. Classical execution times use a different boundary and appear in Table 5.

## 6 Synthetic diagnostic results

The controlled comparison comprises 11,520 scheduled evaluations: twelve configurations on the same 960 synthetic cases, not 11,520 independent graphs. It complements the public construction panel with controlled tests of graph correctness. The full synthetic construction matrix is supplementary (Appendix L). All twelve configurations return valid choices on the 800 query cases. Construction failures and their conditional gaps remain in the separate accounting (Table 18).

**RQ2: Local accuracy does not ensure structural correctness.** Jev answers every adjacency query correctly, but degree accuracy is 61.43%, cycle 66.25% and distance 58.75% (Figure 3). Kev reaches 78.75% adjacency and 19.88% degree; Decider reaches 58.75% and 12.09%. Laya selects **yes** on all 400 binary queries; native-score, mapping and replay checks retain this as observed behavior. Both GPT reasoning references answer all 800 queries correctly.

### Controlled graph-query accuracy

$n = 8, \dots, 12$ ; size-macro accuracy (%); higher is better

| Model                                           | Interface         | Adjacency | Degree | Cycle  | Connectivity | Distance | Articulation |
|-------------------------------------------------|-------------------|-----------|--------|--------|--------------|----------|--------------|
| Jev-1.13.0                                      | native            | 100.00    | 61.43  | 66.25  | 85.00        | 58.75    | 65.00        |
| Qwen3.5-4B                                      | grammar           | 60.00     | 18.32  | 55.00  | 50.00        | 50.00    | 50.00        |
| Qwen3.5-9B                                      | grammar           | 58.75     | 26.99  | 51.25  | 53.75        | 42.50    | 50.00        |
| Qwen3.5-27B                                     | grammar           | 98.75     | 47.70  | 63.75  | 80.00        | 52.50    | 47.50        |
| Qwen3.8-27B                                     | grammar           | 90.00     | 45.50  | 61.25  | 85.00        | 48.75    | 50.00        |
| Qwen2.5-72B-Instruct                            | grammar           | 92.50     | 41.24  | 50.00  | 86.25        | 60.00    | 46.25        |
| Qwen3.5-4B                                      | candidate scoring | 77.50     | 23.89  | 51.25  | 51.25        | 52.50    | 50.00        |
| Decider-2B                                      | native            | 58.75     | 12.09  | 50.00  | 50.00        | 46.25    | 50.00        |
| Kev-4B                                          | native            | 78.75     | 19.88  | 57.50  | 58.75        | 51.25    | 45.00        |
| Laya                                            | native            | 50.00     | 10.87  | 50.00  | 50.00        | 50.00    | 50.00        |
| <b>Separately budgeted reasoning references</b> |                   |           |        |        |              |          |              |
| GPT-5.4                                         | reasoning         | 100.00    | 100.00 | 100.00 | 100.00       | 100.00   | 100.00       |
| GPT-6-Astra                                     | reasoning         | 100.00    | 100.00 | 100.00 | 100.00       | 100.00   | 100.00       |

Accuracy (%)

Figure 3: Taskwise synthetic graph-query accuracy on a common 0–100% scale. Cells show equal-size macro accuracy, not a pooled cross-task score. Degree is multiclass; the other tasks are binary. The direct-decision and separately budgeted reasoning configurations are shown in distinct blocks. Grammar means greedy legal-choice generation. Exact values and macro-versus-pooled definitions are retained in Table 16.

Jev’s point estimates exceed bounded Qwen3.5-27B on all six tasks. The 72B model exceeds Jev on connectivity and distance by just one answer out of eighty each. Thus the queries reveal a capability profile that the adjacency result alone would conceal.

**RQ3: Confirmation and controls.** On the 952-case new-graph confirmation, Jev again has perfect adjacency accuracy, and all four configurations have MaxCut gaps larger than the uniform-policy expectation. Some taskwise rankings change (Appendix P). Presentation controls expose a different issue: option reversal changes every semantic choice on twenty selected Qwen3.5-9B grammar articulation queries, yet accuracy remains 10/20 in both conditions. At an identical token prefix, Qwen3.5-4B candidate-score argmax and masked greedy generation agree on 100/100 states, so argmax alone does not explain the main pipeline’s differences. The separate SemIf device-parity probability check remains unresolved (Appendix C).

Probability quality also differs from answer accuracy [6, 7]: Jev’s above-chance distance accuracy coexists with probability losses worse than uniform on common-valid audit rows. Detailed controls are in Appendix Q. For small constructions, exact best-reachable completions additionally locate irreversible losses along actual histories; this supplementary analysis is confined to the diagnostic bank (Appendix O.3).

## 7 Discussion and limitations

The combined evidence separates three claims that a single success rate would conflate: selecting a legal action, answering a graph query correctly, and completing a good solution. On public instances, complete legal construction coexists with large quality gaps: Jev’s advantage over bounded Qwen on tours does not carry over to signed cuts, and classical methods are stronger on both tasks. Observed responsiveness is therefore not a substitute for graph-solving capability.

A supplementary TSP diagnostic restricts actions to hand-designed append proposals (Appendix H). Gaps decrease, but random proposal selection already has a smaller graph-macro gap than all three full-action model results, and the best fixed rule remains better than every tested model arm. This motivates separate assessment of proposal quality and model selection.

The public and synthetic panels are complementary evaluations, not a controlled size-scaling or size out-of-distribution (OOD) experiment. Synthetic TSP uses Manhattan distance; public TSP uses EUC\_2D and supplies exact current-city distances for every candidate. Public MaxCut uses signed weights on 125-vertex graphs rather than small unweighted graphs. Model rosters and budgets also change: public Qwen3.5-0.8B/2B/4B/9B uses a 16-token ceiling, versus 64 for the earlier Qwen3.5-4B/9B/27B configurations. These public datasets are established benchmarks, not uniformly real-world measurements: the Kro and spin-glass instances are generated.

For the synthetic bank, finite generators, degree conditioning and matching determine the tested distribution. Historical-topology exclusion is local, not proof against pretraining contamination. Later model choices use an inspected bank; only four configurations receive prospective new-graph confirmation. Taskwise estimates do not correct all selection or multiplicity.

Deployment and implementation also differ. Jev is a closed service; trained open readouts use compatible rather than identical upstream environments. The study does not isolate architecture, training method or hardware as a cause of accuracy or latency differences. The synthetic timing panel, size curves, and additional qualifications are retained in Appendices O, K and E. The synthetic LT task is an abstract diffusion objective, not a validated model of human behavior. Local registrations and within-project replay are not third-party attestation.

GraphDecide makes the graph consequences of bounded decisions explicit: judge correctness against the task, judge constructions as complete solutions, and report failures alongside quality. These measurements provide a basis for testing the fast-action promise without assuming that output validity, speed and graph capability are interchangeable.

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## A Synthetic task and interaction contracts

Unless stated otherwise,  $G = (V, E)$  is simple, undirected and unweighted, with integer vertices  $V = \{0, \dots, n-1\}$ , including isolates, and  $m = |E|$  edges. Model-visible inputs contain the graph and the query; action identifiers are strings. Generation families, reference answers, optimum values, and paired-label annotations are not model inputs.

### A.1 Six exact decisions

**Adjacency.** Decide whether  $\{u, v\} \in E$  for distinct supplied vertices.

**Degree.** Return  $d(v)$  from the full choice set  $\{0, \dots, n-1\}$ . The growing output cardinality is an explicit size confound.

**Cycle.** Decide whether any component contains an undirected simple cycle of length at least three.

**Connectivity.** Decide whether a path connects the supplied vertices.

**Distance.** Decide whether the shortest-path distance is at most a supplied threshold  $h$ . Matched challenge pairs have finite distances exactly  $h$  and  $h+1$ , rather than disconnected negative examples.

**Articulation.** Decide whether deleting a supplied vertex increases the total number of connected components, including on disconnected graphs.

### A.2 Four irreversible constructions

**MaxCut.** Vertex 0 is fixed in side A. In increasing vertex order, the model assigns each remaining vertex to A or B, using  $n-1$  decisions. The maximized objective is the number of edges crossing the partition. Empty sides are legal.

**Manhattan TSP.** Distinct integer coordinates define a complete weighted graph with  $d_{ij} = |x_i - x_j| + |y_i - y_j|$ . Starting at 0, the model selects the next unvisited vertex for  $n-2$  decisions. The final unvisited vertex and edge returning to 0 are then forced. This deterministic completion is part of the contract, not answer repair. The minimized objective is full tour length.

**Deterministic normalized linear threshold (LT).** On a directed graph, every incoming edge to  $v$  has weight  $1/d^-(v)$  and the threshold is  $1/2$ , with equality activating. The model selects two distinct seeds, which activate simultaneously at time zero. Subsequent updates are synchronous and irreversible. A zero-indegree nonseed never activates. The maximized objective is final active-set size. This fixed-threshold variant is not the random-threshold classical LT model; we do not import its submodularity or greedy approximation guarantee.

**At-most-two-community modularity.** Using the same fixed-0,  $n-1$  binary assignment protocol, maximize

$$Q(c) = \frac{1}{2m} \sum_{u,v} \left( A_{uv} - \frac{d_u d_v}{2m} \right) \mathbf{1}[c_u = c_v].$$

All-A is a valid one-community solution with  $Q = 0$ . This is a restricted bipartition task, not unrestricted community detection.

**Shared policy.** Every tied optimum is accepted; matching a particular reference trajectory is unnecessary. Models see their own previous choices, not teacher-forced prefixes, except in explicitly separate diagnostic controls. Invalid or incomplete episodes remain infeasible, with no imputed objective, retries, or silent substitution of an action.

## B Controlled data generation and verification

### B.1 Size, structure and pairing

Development covers each integer  $n = 5, \dots, 12$  with two repetitions; the structural challenge covers  $n = 8, \dots, 12$  with eight repetitions. Each size/repetition requests  $n + 14$  instances and  $4n + 8$  maximum calls. Size is a consecutive descriptive axis, not a guarantee of increasing difficulty or a causal intervention holding every other property fixed.

We use heterogeneous block, core-periphery and geometric procedures for query tasks; forests and degree-preserving switches for cycle twins; small-world and switched regular bases for cuts; clustered or perturbed-ring coordinates for TSP; layered directed acyclic graphs for LT; and heterogeneous blocks for modularity. These are finite, overlapping procedures, not uniform samples from all graphs or disjoint mathematical families. Degree conditioning can change a proposal’s original family characteristics.

For adjacency, connectivity and distance, positive and negative queries share a graph and have identical unordered endpoint-degree signatures. Articulation queries share a graph and equal vertex degrees. Cycle twins are different graphs with equal  $n$ ,  $m$  and degree multiset but opposite labels. All five binary tasks are balanced within the audited cells defined by  $n$ ,  $m$ , degree multiset, query degrees and (where applicable)  $h$ , giving a 50% empirical lookup ceiling for those cells. This does not rule out unexamined shortcuts. For example, component count plus  $n, m$  determines undirected cycle existence.

Generation allows at most 128 proposals per cell and records unsupported cells rather than weakening constraints. Four development instances are missing: one cycle pair and one articulation pair at  $n = 5$ . Exhaustion of all 34 unlabeled five-vertex graphs finds one eligible forest/cyclic topology pair but two eligible articulation topologies; the latter shortfall is therefore a generator-support/budget limitation, not a theorem of impossibility. The original 960-instance challenge has no omissions.

### B.2 Identity exclusion and evidence chronology

The original challenge excludes an audited pool of 1,040 exact-labeled deduplicated records from 121 earlier artifacts. Fingerprints select candidate buckets; the VF2 graph-isomorphism algorithm determines identity exactly. TSP uses complete weighted distance graphs, and LT uses directed topology. Independent checks found no prior overlap and no same-task duplicate across nominally independent clusters. The 1,040 records are not claimed to be 1,040 isomorphism classes.

Crucially, the common-interface matrix in this paper reuses these already inspected banks. Its protocol was frozen before its own calls, but it is *not a newly blind test*. Earlier development diagnosed wording and conformance issues. A subsequent prospective confirmation excludes these banks and retains its own results, including missing cells, rather than replacing them after inspecting outcomes (Section P). We preserve the chronological distinction between original strict experiments, diagnostic controls and the revised common-interface matrix.

Table 1: Dataset counts and maximum calls. An additional forty weighted TSP instances support wording confirmation and bring the synthetic collection to 1,356. They are not additional rows of the canonical comparison.

| Task             | Development | Challenge | Decisions per instance |
|------------------|-------------|-----------|------------------------|
| Adjacency        | 32          | 80        | 1                      |
| Degree           | 136         | 400       | 1                      |
| Cycle            | 30          | 80        | 1                      |
| Connectivity     | 32          | 80        | 1                      |
| Distance         | 32          | 80        | 1                      |
| Articulation     | 30          | 80        | 1                      |
| MaxCut           | 16          | 40        | $n - 1$                |
| TSP              | 16          | 40        | $n - 2$                |
| Deterministic LT | 16          | 40        | 2                      |
| Modularity       | 16          | 40        | $n - 1$                |
| Total            | 356         | 960       | 668 / 1,920 per model  |

**Meaning of registration and independent checking.** Here, “registered” or “preregistered” means a locally persisted design record frozen before the specified calls (and, for the confirmation, before generation), not a deposit in an external public preregistration registry. Source and artifact hashes detect changes relative to those records; they are not third-party timestamps or cryptographic proof of the full chronology. Independent verification means separate replay or mathematical computation within this project, not replication by an external research group.

### B.3 Independent truth and artifact boundaries

New adjacency, distance and articulation oracles were exhaustively checked on all 1,024 labeled five-vertex graphs. Every released label, feasible reference, baseline and optimum passes independent validation; TSP uses exact subset dynamic programming and an independent optimum check, while the small partition and seed spaces permit exhaustive verification.

The synthetic dataset bundle separates five model-visible fields (identifier, task, kind, state, query budget), private references, and analysis metadata. Its request builder does not read references. Offline scoring is deliberately privileged: giving that scorer to a model would change the benchmark. This dataset bundle, distinct from the manuscript source archive, was extracted into an isolated Python process and all 1,356 instances were verified without importing model clients or inference runtimes.

## C Synthetic evaluation design

### C.1 Evidence sets and permissible comparisons

Table 2 distinguishes the complete synthetic comparison from its four-configuration diagnostic and confirmation subsets. Reusing graph instances allows paired comparisons but does not add independent test graphs. Freezing a later model’s protocol before its calls does not make an already inspected bank a blind holdout.

Table 2: Synthetic evidence sets, not a sum of independent graphs. Later presentation, probability and deadline controls reuse their explicitly named banks. Cases are scheduled episodes, not necessarily completed ones.

| Evidence set                       | Cases per method | Interpretation                                                                                                                                                      |
|------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Original four-configuration matrix | 1,316            | 356 development and 960 inspected challenge cases.                                                                                                                  |
| Prospective confirmation           | 952              | Same four configurations; new seed and prior-topology exclusion, but unchanged generator procedures and sizes.                                                      |
| Canonical GPT-5.4                  | 960              | Reuses the inspected challenge; identical initial logical requests, not matched reasoning or compute.                                                               |
| Full cross-family matrix           | 960              | Twelve configurations on that same challenge, including canonical GPT-6-Astra; ten direct-decision configurations and two separately labeled GPT reasoning anchors. |
| Larger/newer local extension       | 960              | Qwen3.8-27B and Qwen2.5-72B-Instruct, included in the twelve above; not additional graphs or a scale-only ablation.                                                 |
| Legacy GPT-6-Astra                 | 960              | Same challenge graphs, different instructions and option descriptions; separate protocol.                                                                           |
| Expanded local/readout panel       | 64               | Reuses selected confirmation cases at $n = 8, 12$ ; exploratory, not a new blind replication.                                                                       |

## C.2 Completed common-interface configurations

Every method receives the same initial logical request, including explicit action descriptions. Questions specify mathematics rather than requiring a wire encoding that could conflict with a letter-based readout. TSP explicitly names the current endpoint and actual visited prefix.

**Jev action.** The canonical synthetic matrix uses its own registered action-eligibility policy, distinct from earlier strict-response controls and the later public-instance protocol. Expected model, response shape, legal choice, probability keys and finite ranges, confidence and usage must still validate. Probability normalization and choice/argmax disagreement are separately recorded observations, not reasons to replace the supplied action.

**4B and 9B grammar.** Qwen3.5-4B and Qwen3.5-9B, greedy no-thinking generation, 64-token ceiling and legal-choice grammar, with the existing output parser. Thinking-enabled extensions are reported separately.

**4B scores.** The existing 4B checkpoint with pinned SemIf-style rendering and exact requested candidate-token log probabilities from vLLM. The chosen action is the conditional argmax over *all* candidates, not a natural top- $K$  approximation. One token generated by the application programming interface (API) is unused for selection and is counted in resource accounting.

**27B grammar.** Qwen3.5-27B with the same no-thinking, 64-token, legal-choice grammar policy and a 180-second request deadline.

**Larger/newer local grammar.** Qwen3.8-27B and Qwen2.5-72B-Instruct use full BF16 weights, greedy legal-choice grammar, a 64-token ceiling and a 180-second deadline. Thinking is explicitly disabled for Qwen3.8; Qwen2.5 uses its standard instruct template without added

reasoning. They use two and four RTX A6000 graphics processing units (GPUs), respectively. The 72B checkpoint is an older-generation dense model, not a scale-only variant of Qwen3.5/3.8.

**Trained Jev-like readouts.** Decider-2B v11 and Kev-4B use their pinned upstream native inference paths; Laya uses its encoder and trained choice head on a central processing unit (CPU), using 32-bit floating point (FP32). Lossless context checks remain binding. These are compatible-runtime evaluations, not certifications of identical upstream environments.

**Reasoning anchors.** GPT-5.4 and GPT-6-Astra use tool-free, default-reasoning clients with 4,096 output tokens and a 180-second deadline. They are separated from the ten direct-decision configurations throughout.

No Jev outputs are used to train or distill a replacement model. Local models run on RTX A6000 GPUs; concurrent generation and scoring requests to the shared Qwen3.5-4B endpoint are serialized. The checkpoint, renderer, environment and per-call settings are recorded. This controls some factors, but the 4B readout comparison also changes rendering and cannot identify a pure causal effect of the scoring rule.

**A failed reproduction check is retained.** On twelve matched CPU SemIf probes, both GPU scoring routes matched every chosen action. However, maximum probability errors were 0.0166305 for requested token scores and 0.0379656 for teacher forcing, both exceeding the pre-frozen 0.01 threshold; the routes differed by 0.0213351. Independent token indexing and high-precision softmax recomputation found no parser mismatch. The numerical cause remains unresolved. Thus “4B scores” is an independent reference using SemIf rendering, not certified numerical reproduction of CPU SemIf, and the failed gate is not relaxed.

### C.3 Metrics and statistical units

For a synthetic exact task, the primary accuracy is the mean of the size-specific accuracies, with invalid/unattempted scheduled cases retained in denominators. For degree, macro-size and pooled accuracy differ because there are  $n$  classes per size. The synthetic construction-quality metric is mean additive objective gap over feasible completed episodes, accompanied by feasible coverage over all scheduled episodes. Optimum-hit counts are supplementary diagnostics, not the primary comparison or a substitute for graded quality. For maximization gap =  $f^* - f$ , and for minimization gap =  $f - f^*$ . The mean averages per-episode gaps, not graph sizes or success indicators. Lower is better, and zero means optimal on the evaluated feasible cases. The synthetic construction tables use additive gaps: edges for MaxCut, distance units for TSP, activated vertices for LT, and modularity units for community partitioning. The separately recorded TSP relative gap divides each episode’s gap by its positive optimum; an average relative gap must not be inferred by dividing aggregate means. Modularity ratios are not used. Different task units are never pooled into one quality score.

Generation omissions and model failures have different denominators. An ungenerated case is a coverage shortfall, with no model prediction to score; confirmation accuracy is conditional on the 952 realized cases, not all 960 requested generation slots. Once an instance is generated and scheduled for a method, unsupported input, timeout, or invalid output remains in that method’s scheduled denominator. Such failures do not receive a fabricated objective. Conditional gap zero means only that all completed solutions are optimal; it does not imply perfect scheduled performance.

For the original four-configuration evidence, uncertainty is estimated by 2,000 paired bootstrap replicates, stratified by size. The full twelve-configuration tables report descriptive point estimates, not new all-pairs significance tests. Resampling units are the transitive union of graph identity and

paired construction relations: cycle twins must remain together even though they have different graphs. The reported intervals are exploratory pointwise percentile intervals conditional on these generators and samples. They are not family-wise significance tests; degenerate intervals do not establish equivalence or population certainty.

**Classical and uniform references.** We include greedy cut plus best-improvement vertex flips, nearest-neighbor TSP plus best-improvement 2-opt, deterministic-LT marginal greedy, and deterministic multistart single-vertex modularity search. These methods can revise choices and are quality anchors, not equal-budget online methods. For each task we additionally compute the exact expected objective under uniform legal sequential actions, rather than relying on one random rollout. An all-A, strict-positive-improvement modularity sweep is a weak reference that remains at zero; it is not our sole classical comparator.

## D Synthetic coverage and evidence integrity

**Complete accounting, including unsupported inputs.** Qwen3.5-27B, Qwen3.8-27B, Qwen2.5-72B, Decider and Kev each complete 960 episodes in 1,920 calls without execution failures. Laya completes 951 episodes; nine initial LT inputs fail its lossless native context/head limits. Its ledger has 1,911 entries but only 1,902 model forwards: pre-inference rejections are not credited as inference. GPT-5.4 retains its nine construction timeouts, while canonical GPT-6-Astra completes all 960 episodes in 1,920 calls. No failed or unsupported case is replaced, retried, truncated, or removed from scheduled denominators (Table 18).

**Evidence integrity.** The original combined offline exporter admits only verified terminal real-model lanes, reconstructs 9,600 episode records, and independently checks every feasible construction objective. A versioned publication export adds the two completed BF16 lanes, whose 1,920 episodes and 320 constructions have separate runner verification and independent mathematical replay. The combined presentation contains 720 task/size/configuration cells, preserving the original 600. No new model calls are made during export. Compact tables are generated from the sealed task/size metrics and retain macro versus pooled exact accuracy and conditional versus scheduled outcomes. Recorded Python-version aggregate roundoff is handled separately from exact request, action, episode and count checks. These are within-project verification procedures, not third-party attestation. The larger-local extension also checks served tokenization against pinned nonthinking rendering. No failed answer is rerun to improve the reported result.

## E Limitations and responsible use

**Diagnostic scale and distribution.** The synthetic diagnostic graphs have at most twelve vertices. This enables exhaustive validation and continuation analysis but does not establish performance at operational scale. Consecutive sizes are neither a size-OOD test nor a universal difficulty ordering. Finite generator support, degree conditioning, and matched-pair selection change the evaluated distribution. Topology exclusion is relative to an auditable local history, not proof of absence from pretraining.

**Model and interface coverage.** Jev is a closed service; no architecture or training attribution can be independently established here. No-thinking Qwen is a bounded baseline, not its full reasoning capacity. The Laya, Decider and Kev extensions test open trained readouts, not reproductions of

Jev’s undisclosed architecture. Decider/Kev are compatible-runtime upstream-inference evaluations, not complete upstream-environment reproductions or broad CPU/GPU numerical-parity claims. Fully matched rendering across readouts and deployment-matched latency experiments remain distinct research questions. The earlier SemIf candidate-token CPU/GPU numerical reproduction failure is explicitly unresolved.

**Inference and statistical uncertainty.** The synthetic common-interface matrix follows inspected development evidence. Small numbers of construction clusters, multiple task comparisons and API variability limit inference. Pointwise bootstrap intervals do not correct selection or multiplicity, and no overall “best model” score is defined. Repeated requests and option ordering receive diagnostic, not exhaustive, coverage. The prospective replication is complete. The sealed canonical strong-model anchor is reported separately with its own reasoning and serving conditions; it is not a matched-compute comparison. Only the original four configurations receive the prospective new-graph confirmation. GPT, the larger-local and trained-readout extensions therefore do not establish replicated rankings on fresh graphs. Public-instance construction is a separate evaluation panel, not a replication of the synthetic query results. Unseen generator families, additional independent seeds and specialized graph-trained policies would test further aspects of external validity.

**Artifact and societal scope.** The diagnostic-suite prompts contain synthetic graphs, not personal-network records. The LT task is an abstract diffusion objective; it does not validate manipulation of people or predict real social behavior.

## F Public-instance contracts and validation

This panel is separate from the 960-case synthetic matrix. The following contracts govern its instances, interactions and scoring.

**TSP instances and distances.** Seven TSPLIB instances [23] contain 100, 100, 100, 127, 150, 200 and 225 cities. Their native edge-weight type is `EUC_2D`:

$$d_{ij} = \left\lceil \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} + \frac{1}{2} \right\rceil.$$

Thus evaluation uses nearest-integer Euclidean distances, not the Manhattan metric of the synthetic TSP bank or unrounded coordinate distances. For this panel, coordinate differences, products and the sum inside the square root are evaluated with ordinary double-precision floating-point operations before rounding. Exact-decimal evaluation and alternative `hypot` arithmetic are not substituted for this source convention. Published proven optimum values provide the references. A complete tour visits every city once and returns to the starting city. The full instance is supplied; every unvisited city is a legal next action, without nearest-neighbor candidate filtering. The model-visible state also supplies the exact current-city distance for every candidate under the instance’s edge-weight convention. The final remaining city and return edge are forced under the construction contract.

**Signed public cut objective.** The ten OPTSICOM Set2 instances [17, 19] are three-dimensional spin-glass graphs with  $n = 125$ ,  $m = 375$ , and edge weights  $w_{uv} \in \{-1, +1\}$ . For a partition  $(A, V \setminus A)$ , the maximized objective is

$$f(A) = \sum_{\{u,v\} \in E} w_{uv} \mathbf{1}[\{u,v\} \cap A = 1].$$

Crossing a negative edge reduces this objective. Counting crossing edges or taking absolute weights would therefore evaluate a different problem. The reference is the source-reported historical best-known cut value, from a mirror pinned to August 16, 2016 and retrieved September 29, 2026 [17]. It is not asserted to be a current global record or a proven optimum. The relative reference gap is  $100(r - f(A))/r$ , with  $r > 0$ . A negative value is retained and denotes improvement over this historical reference; values above 100% denote a negative achieved cut objective. Exact-filename reference attribution uses the mirror’s explicit best-known-value records. The primary OPTSICOM table uses different labels; we do not assume an independently established alias mapping between them.

**Configurations and pairing.** The direct-decision panel comprises Jev and no-thinking Qwen3.5-0.8B, Qwen3.5-2B, Qwen3.5-4B and Qwen3.5-9B grammar generation. The local grammar ceiling is 16 tokens, not the synthetic matrix’s 64-token ceiling. Three relabeling/start conditions are paired across these configurations. The schedule therefore contains 51 episodes per configuration: 21 tours and 30 cuts, all completed. Each model permits at most four concurrent episodes. The three conditions reuse each underlying instance; they are not three independently sampled graph sets. Jev uses version 1.13.0 and a 30-second per-call deadline; the Qwen configurations use greedy decoding and a 180-second per-call deadline. Each configuration makes 6,684 evaluated decisions: 2,964 on tours and 3,720 on cuts. The admitted panel contains 255 complete episodes.

**Calibration and action eligibility.** Before the public schedule, three unscored calibration trajectories on `eil51` each encountered a numeric probability vector summing to 0.99, with probabilities on a 0.01 grid. The strict calibration adapter rejected the vector and invalidated the trajectory. These calibration trajectories and the `eil51` instance are excluded from benchmark results; they are not failures silently removed from the public schedule.

The subsequently approved public protocol validates and scores Jev’s legal native selected action independently of probability-vector conformance. Numeric probability vectors, their sums and conformance diagnostics are recorded separately. A malformed vector is not normalized, and its argmax does not replace the selected action. There is no retry to obtain a better action or a conforming vector. Invalid actions and execution failures remain failures under scheduled accounting. This separates construction performance from probability reporting; it does not certify the latter as calibrated or mathematically valid. The earlier synthetic scores and protocols are unchanged. A subsequent native-action calibration completed all 49 `eil51` decisions with two probability-audit failures and no action failures. It also remains excluded from the benchmark panel; completion on this calibration instance is not a public-suite performance result.

**Public probability reports.** Of Jev’s 6,684 evaluated calls, 5,497 have conformant probability reports, 1,186 have nonunit sums and one has a selected-choice/probability mismatch. All selected actions remain legal and are scored under this public panel’s native-action contract. Complete construction therefore does not imply valid probability reporting.

**Outcome admission.** All 255 model objectives and 153 classical objectives were independently recomputed. Tour checks enforce a full permutation and closure, then sum the native distances. Cut checks enforce a complete partition and compare the signed objective with NetworkX weighted cut size. All actual requests and retained trajectories were replayed under the final distance convention, rather than checking final objectives alone. The classical random reference is one sampled legal

solution per instance and condition, not the exact uniform-policy expectation used in some synthetic diagnostics.

**Rounding correction audit.** The initial public TSP implementation used `hypot` rather than the source’s double-precision square-root expression. On `tsp225`, the two computations assign weights 142 and 143, respectively, to the original one-based edge 75–111. The conventional square-root calculation reproduces the published 3,916-length optimum tour. Exact-decimal reinterpretation instead yields 3,921 for that tour and is not the source convention. The official tour does not traverse the discrepant edge, so that tour alone does not establish equivalence of the two floating-point matrices. After correction, all three `tsp225` conditions were rerun in full for every model; the fifteen superseded episodes do not enter the results. The other sixteen instances were retained only after complete actual-request and score replay under the corrected convention. This uniform replacement rule does not select episodes by solution quality.

## G Public-instance references and detailed outcomes

Table 3: Public-instance mean relative gap and complete coverage. Lower gaps are better; cut gaps use historical references, not proven optima. Each instance has three conditions. The classical random row is sampled, not an exact policy expectation. The 2-opt reference starts from nearest neighbor; single-flip search starts from incremental greedy.

| Method                                | TSP gap (%) | Coverage | MaxCut gap (%) | Coverage |
|---------------------------------------|-------------|----------|----------------|----------|
| Jev (native action)                   | 66.01       | 21/21    | 104.26         | 30/30    |
| Qwen3.5-0.8B                          | 750.53      | 21/21    | 100.64         | 30/30    |
| Qwen3.5-2B                            | 715.96      | 21/21    | 100.23         | 30/30    |
| Qwen3.5-4B                            | 307.11      | 21/21    | 100.05         | 30/30    |
| Qwen3.5-9B                            | 381.82      | 21/21    | 97.31          | 30/30    |
| Classical references (all conditions) |             |          |                |          |
| Uniform random                        | 772.66      | 21/21    | 100.28         | 30/30    |
| Nearest neighbor / incremental greedy | 27.65       | 21/21    | 29.20          | 30/30    |
| 2-opt / single-flip search            | 5.93        | 21/21    | 22.02          | 30/30    |

Table 4 identifies all seventeen benchmark instances. TSPLIB’s published optimum list supplies the tour references [23]; these are source-reported proven optima, not optimality proofs newly established here. The signed-cut references use the exact instance filenames in the OPTSI-COM mirror’s best-known-value records [17, 19]. The pinned revision is `9c9c1a84774ac14b7c161d5d5ea1f4f554a39a76`, dated August 16, 2016; source retrieval was September 29, 2026. Verified feasible witnesses establish attainability of those cut values, not their current or proven optimality. No alias correspondence to the primary site’s differently named rows is assumed.

**Per-instance outcomes.** Tables 6 and 7 report every model objective, without averaging away the three paired conditions  $c \in \{0, 1, 2\}$ . All listed episodes complete. The reference column is repeated to make each row independently interpretable. Tables 8 and 9 give the corresponding classical objectives. Nearest neighbor supplies the starting tour for 2-opt; incremental greedy supplies the starting cut for single-flip improvement. These starts and subsequent revisions are part of each classical method. Negative signed-cut objectives are preserved.

Table 4: Public instance inventory and scoring references. TSP references are published proven optima; signed MaxCut references are historical source-reported best-known values. The calibration instance `eil151` is excluded. TSP edge counts describe the full undirected distance graph.

| Instance                  | Vertices | Edges | Reference |
|---------------------------|----------|-------|-----------|
| <b>TSP</b>                |          |       |           |
| <code>kroA100</code>      | 100      | 4950  | 21282     |
| <code>kroB100</code>      | 100      | 4950  | 22141     |
| <code>kroC100</code>      | 100      | 4950  | 20749     |
| <code>bier127</code>      | 127      | 8001  | 118282    |
| <code>kroA150</code>      | 150      | 11175 | 26524     |
| <code>kroB200</code>      | 200      | 19900 | 29437     |
| <code>tsp225</code>       | 225      | 25200 | 3916      |
| <b>Signed MaxCut</b>      |          |       |           |
| <code>sg3dl051000</code>  | 125      | 375   | 110       |
| <code>sg3dl052000</code>  | 125      | 375   | 112       |
| <code>sg3dl053000</code>  | 125      | 375   | 106       |
| <code>sg3dl054000</code>  | 125      | 375   | 114       |
| <code>sg3dl055000</code>  | 125      | 375   | 112       |
| <code>sg3dl056000</code>  | 125      | 375   | 110       |
| <code>sg3dl057000</code>  | 125      | 375   | 112       |
| <code>sg3dl058000</code>  | 125      | 375   | 108       |
| <code>sg3dl059000</code>  | 125      | 375   | 110       |
| <code>sg3dl0510000</code> | 125      | 375   | 112       |

Table 5: Median times in seconds. Models: audited episode wall time, including the evaluation overhead specified in Section 5. Classical methods: recorded local execution time. Paired method labels denote TSP / MaxCut. Timing boundaries and compute budgets are not matched.

| Method                                | TSP seconds | MaxCut seconds |
|---------------------------------------|-------------|----------------|
| Jev (native action)                   | 20.95       | 21.25          |
| Qwen3.5-0.8B                          | 48.76       | 33.52          |
| Qwen3.5-2B                            | 55.59       | 35.43          |
| Qwen3.5-4B                            | 103.49      | 49.45          |
| Qwen3.5-9B                            | 154.56      | 71.29          |
| Sampled random                        | 0.004103    | 0.000967       |
| Nearest neighbor / incremental greedy | 0.009168    | 0.001111       |
| NN + 2-opt / greedy + single flip     | 0.035151    | 0.001306       |

Table 6: Public TSP tour lengths for every model and condition  $c$ . Lower is better. References are published proven optima. Columns retain full model identities; all Qwen configurations use grammar generation with thinking disabled.

| Instance | $c$ | Reference | Jev    | Qwen3.5-<br>0.8B | Qwen3.5-<br>2B | Qwen3.5-<br>4B | Qwen3.5-<br>9B |
|----------|-----|-----------|--------|------------------|----------------|----------------|----------------|
| kroA100  | 0   | 21282     | 33135  | 168766           | 159181         | 64364          | 103906         |
| kroA100  | 1   | 21282     | 28231  | 158474           | 157078         | 65182          | 71947          |
| kroA100  | 2   | 21282     | 33565  | 159436           | 144143         | 70365          | 75285          |
| kroB100  | 0   | 22141     | 34488  | 174458           | 172522         | 67883          | 86406          |
| kroB100  | 1   | 22141     | 41909  | 166630           | 160250         | 61365          | 56516          |
| kroB100  | 2   | 22141     | 34679  | 156383           | 146990         | 66345          | 101084         |
| kroC100  | 0   | 20749     | 30220  | 188589           | 177320         | 64093          | 99987          |
| kroC100  | 1   | 20749     | 38357  | 162326           | 159382         | 58250          | 83141          |
| kroC100  | 2   | 20749     | 31069  | 159361           | 153745         | 66868          | 76161          |
| bier127  | 0   | 118282    | 185736 | 653563           | 621113         | 363014         | 405887         |
| bier127  | 1   | 118282    | 172050 | 628031           | 618998         | 367045         | 445228         |
| bier127  | 2   | 118282    | 192592 | 650084           | 616121         | 350059         | 399125         |
| kroA150  | 0   | 26524     | 45165  | 254544           | 233344         | 137385         | 151954         |
| kroA150  | 1   | 26524     | 43457  | 241820           | 228605         | 127883         | 146992         |
| kroA150  | 2   | 26524     | 40795  | 233135           | 230406         | 126243         | 153664         |
| kroB200  | 0   | 29437     | 42085  | 345722           | 333674         | 157237         | 110990         |
| kroB200  | 1   | 29437     | 57879  | 335113           | 311582         | 156568         | 172921         |
| kroB200  | 2   | 29437     | 50857  | 329912           | 294366         | 158194         | 143201         |
| tsp225   | 0   | 3916      | 8566   | 39194            | 39938          | 26398          | 30617          |
| tsp225   | 1   | 3916      | 7529   | 39048            | 40308          | 23511          | 33877          |
| tsp225   | 2   | 3916      | 7116   | 41244            | 39965          | 21445          | 28399          |

Table 7: Public signed-MaxCut objectives for every model and condition  $c$ . Higher is better. The reference is historical and is not a proven optimum. Negative objectives, and hence relative gaps above 100%, are valid results.

| Instance     | $c$ | Reference | Jev | Qwen3.5-<br>0.8B | Qwen3.5-<br>2B | Qwen3.5-<br>4B | Qwen3.5-<br>9B |
|--------------|-----|-----------|-----|------------------|----------------|----------------|----------------|
| sg3d1051000  | 0   | 110       | -4  | -2               | -4             | -4             | -2             |
| sg3d1051000  | 1   | 110       | -2  | 2                | -2             | 4              | 14             |
| sg3d1051000  | 2   | 110       | 0   | 4                | 0              | 0              | 18             |
| sg3d1052000  | 0   | 112       | -6  | 4                | 4              | 4              | -2             |
| sg3d1052000  | 1   | 112       | 2   | 4                | -2             | -2             | 2              |
| sg3d1052000  | 2   | 112       | 2   | -6               | 2              | 2              | 4              |
| sg3d1053000  | 0   | 106       | -2  | 2                | 2              | 2              | 14             |
| sg3d1053000  | 1   | 106       | -10 | -2               | 0              | 0              | -10            |
| sg3d1053000  | 2   | 106       | 0   | -2               | 0              | 0              | -4             |
| sg3d1054000  | 0   | 114       | -12 | -8               | -4             | -4             | 0              |
| sg3d1054000  | 1   | 114       | -4  | -4               | -4             | -4             | -26            |
| sg3d1054000  | 2   | 114       | -24 | 2                | 0              | 0              | 10             |
| sg3d1055000  | 0   | 112       | -2  | -4               | -2             | -2             | 14             |
| sg3d1055000  | 1   | 112       | -8  | -8               | -2             | -2             | 10             |
| sg3d1055000  | 2   | 112       | -2  | -6               | -2             | -2             | 8              |
| sg3d1056000  | 0   | 110       | 2   | -4               | 2              | 2              | 2              |
| sg3d1056000  | 1   | 110       | 6   | 2                | 0              | 0              | -6             |
| sg3d1056000  | 2   | 110       | 0   | -4               | -2             | -2             | 8              |
| sg3d1057000  | 0   | 112       | -16 | 4                | 2              | 2              | 12             |
| sg3d1057000  | 1   | 112       | 0   | -10              | -2             | -2             | 18             |
| sg3d1057000  | 2   | 112       | 8   | 4                | -2             | -2             | 4              |
| sg3d1058000  | 0   | 108       | -10 | 2                | 2              | 2              | -14            |
| sg3d1058000  | 1   | 108       | -12 | 4                | 0              | 0              | 2              |
| sg3d1058000  | 2   | 108       | -6  | 4                | 0              | 0              | -8             |
| sg3d1059000  | 0   | 110       | -6  | -6               | 2              | 2              | 8              |
| sg3d1059000  | 1   | 110       | -4  | 12               | 6              | 6              | 2              |
| sg3d1059000  | 2   | 110       | -2  | -6               | 0              | 0              | 8              |
| sg3d10510000 | 0   | 112       | -2  | -6               | 2              | 2              | -8             |
| sg3d10510000 | 1   | 112       | -14 | 0                | -4             | -4             | 0              |
| sg3d10510000 | 2   | 112       | -14 | 6                | 0              | 0              | 12             |

Table 8: Public TSP classical tour lengths under the same three conditions  $c$  per instance. Lower is better. The sampled random policy is evaluated once per condition; NN + 2-opt includes its nearest-neighbor initialization.

| Instance | $c$ | Reference | Sampled random | Nearest neighbor | NN + 2-opt |
|----------|-----|-----------|----------------|------------------|------------|
| kroA100  | 0   | 21282     | 157320         | 27904            | 23246      |
| kroA100  | 1   | 21282     | 184656         | 26309            | 22088      |
| kroA100  | 2   | 21282     | 173913         | 28504            | 22137      |
| kroB100  | 0   | 22141     | 174804         | 28478            | 22778      |
| kroB100  | 1   | 22141     | 159032         | 29365            | 23349      |
| kroB100  | 2   | 22141     | 161332         | 28343            | 23058      |
| kroC100  | 0   | 20749     | 178066         | 25635            | 21747      |
| kroC100  | 1   | 20749     | 162099         | 27489            | 22002      |
| kroC100  | 2   | 20749     | 174211         | 24857            | 22409      |
| bier127  | 0   | 118282    | 662874         | 137429           | 123363     |
| bier127  | 1   | 118282    | 661530         | 147899           | 123178     |
| bier127  | 2   | 118282    | 630279         | 148831           | 125505     |
| kroA150  | 0   | 26524     | 263824         | 35174            | 29048      |
| kroA150  | 1   | 26524     | 267379         | 32959            | 27979      |
| kroA150  | 2   | 26524     | 250112         | 34094            | 27870      |
| kroB200  | 0   | 29437     | 312060         | 38612            | 31214      |
| kroB200  | 1   | 29437     | 342736         | 36969            | 31971      |
| kroB200  | 2   | 29437     | 346746         | 39006            | 32053      |
| tsp225   | 0   | 3916      | 43191          | 4929             | 4186       |
| tsp225   | 1   | 3916      | 39390          | 4864             | 4031       |
| tsp225   | 2   | 3916      | 42174          | 5286             | 4239       |

Table 9: Public signed-MaxCut classical objectives under the same three conditions  $c$  per instance. Higher is better. Single-flip search includes its greedy initialization. These are achieved objectives, not claims of optimality or new best-known records.

| Instance     | $c$ | Reference | Sampled random | Greedy | Greedy + single flip |
|--------------|-----|-----------|----------------|--------|----------------------|
| sg3d1051000  | 0   | 110       | -2             | 84     | 86                   |
| sg3d1051000  | 1   | 110       | -8             | 72     | 82                   |
| sg3d1051000  | 2   | 110       | 0              | 70     | 86                   |
| sg3d1052000  | 0   | 112       | -4             | 70     | 72                   |
| sg3d1052000  | 1   | 112       | 2              | 68     | 78                   |
| sg3d1052000  | 2   | 112       | 10             | 80     | 94                   |
| sg3d1053000  | 0   | 106       | -4             | 76     | 84                   |
| sg3d1053000  | 1   | 106       | -2             | 78     | 90                   |
| sg3d1053000  | 2   | 106       | -8             | 86     | 92                   |
| sg3d1054000  | 0   | 114       | -8             | 72     | 82                   |
| sg3d1054000  | 1   | 114       | -14            | 82     | 94                   |
| sg3d1054000  | 2   | 114       | 2              | 78     | 88                   |
| sg3d1055000  | 0   | 112       | 6              | 76     | 86                   |
| sg3d1055000  | 1   | 112       | 18             | 78     | 88                   |
| sg3d1055000  | 2   | 112       | 4              | 88     | 90                   |
| sg3d1056000  | 0   | 110       | 8              | 84     | 84                   |
| sg3d1056000  | 1   | 110       | 0              | 76     | 86                   |
| sg3d1056000  | 2   | 110       | -8             | 74     | 88                   |
| sg3d1057000  | 0   | 112       | 4              | 84     | 88                   |
| sg3d1057000  | 1   | 112       | 0              | 80     | 86                   |
| sg3d1057000  | 2   | 112       | -10            | 86     | 88                   |
| sg3d1058000  | 0   | 108       | -16            | 78     | 86                   |
| sg3d1058000  | 1   | 108       | 0              | 78     | 84                   |
| sg3d1058000  | 2   | 108       | -6             | 84     | 94                   |
| sg3d1059000  | 0   | 110       | -8             | 70     | 84                   |
| sg3d1059000  | 1   | 110       | 6              | 88     | 92                   |
| sg3d1059000  | 2   | 110       | 4              | 64     | 76                   |
| sg3d10510000 | 0   | 112       | 12             | 82     | 84                   |
| sg3d10510000 | 1   | 112       | 8              | 86     | 92                   |
| sg3d10510000 | 2   | 112       | 6              | 76     | 82                   |

## H Supplementary TSP action-abstraction diagnostic

This diagnostic asks how much of the full-action TSP gap can be reduced by offering hand-designed next-city proposals, and whether naming the rules adds value beyond anonymous proposals. It reuses the same seven TSPLIB instances and three relabeling/start conditions as the primary public panel, for Jev, Qwen3.5-4B and Qwen3.5-9B. It is not a new benchmark or an adaptive search for a successful inference configuration.

### H.1 Frozen design and append-only rules

The protocol, four rule formulas, model settings, random seeds, analysis and inclusion criteria were frozen before any new B/C inference. The earlier A results were already known: this registration was not blind to full-action performance. The prespecified appendix-inclusion criterion required independent verification and at least 95% completion in every model/arm. The additional descriptive improvement criterion required a graph-macro gain of at least one percentage point and improvement on at least five of seven graphs, with all models and contrasts disclosed. These criteria are reporting rules, not statistical significance tests.

**A: full actions.** Reuse the 63 verified primary trajectories for the three models, including all corrected `tsp225` replacements. Every unvisited city is an available next action.

**B: anonymous proposals.** Offer the distinct cities proposed by the four frozen rules. The full graph, actual prefix and shared numeric candidate features remain visible, without rule names or rule-to-city mapping.

**C: named proposals.** At the same history, use the same proposed cities, numeric features and option order as B, adding rule names, formulas and the mapping from rules to their proposed cities.

Let  $c$  be the current city,  $s$  the start city (ID 0 under the relabeling), and  $U$  all unvisited cities. While  $|U| \geq 2$ , define

$$e(v) = d(c, v), \quad h(v) = \min_{w \in U \setminus \{v\}} d(v, w), \quad r(v) = d(v, s).$$

Each rule proposes the city in *all* of  $U$  minimizing its score (Table 10), breaking ties by lowest relabeled city ID. Coefficients are fixed, not fitted. The model’s action appends one city; it does not invoke an insertion operator, local search, or a multi-step solver. The graph itself is not cropped, and distances from the current city to all legal cities remain available even though B/C restrict the offered actions.

Table 10: Frozen append-proposal rules. Each score is minimized over all unvisited cities;  $h$  is computed using the full remaining set, not just other proposed cities. Rule names describe the fixed formulas, not guarantees.

| Rule               | Score          |
|--------------------|----------------|
| Nearest            | $e(v)$         |
| One-step lookahead | $e(v) + h(v)$  |
| Return reserve     | $2e(v) - r(v)$ |
| Isolation priority | $e(v) - h(v)$  |

Rules proposing the same city are grouped into one option, so a city does not receive extra probability mass merely because several rules propose it. A single distinct proposal is appended deterministically without a model call. The final city and return edge are also forced, as in the primary contract. The sorted distinct cities are shuffled using a local history-dependent seed: the hash of the frozen seed, instance identifier and selected-city history. Thus B/C ordering agrees at matched histories. Their trajectories may subsequently diverge, and C also changes semantic content and prompt length; this is not a pure causal isolation of planning ability.

Jev uses the public native-action policy. The Qwen configurations retain greedy, no-thinking grammar generation with a 16-token ceiling. Per-call adapter deadlines remain 30 seconds for Jev and 180 seconds for Qwen, with a separate watchdog five seconds longer and a 900-second episode solve deadline. At most four episodes per model run concurrently across B and C. No failed action is retried, repaired or replaced.

## H.2 Coverage, quality and paired effects

The schedule contains 126 new B/C episodes, of which 125 complete. The sole failure is Jev B on kroB200, condition 1: after two forced proposal steps, the first model call returns Hypertext Transfer Protocol (HTTP) status 520. The episode remains incomplete with no objective or imputed gap. Jev B therefore completes 20/21 cases (95.24%); every other model/arm completes 21/21. All satisfy the frozen coverage criterion. Four successful new Jev calls have nonunit probability sums, two in each arm. Their native actions are retained without normalization or argmax substitution; probability conformance is separate from action eligibility.

Gaps retain the primary TSP definition,  $100(f - f^*)/f^*$ . For each model/arm, first average completed conditions within each graph, then average the seven graphs equally. Coverage always uses all scheduled conditions. Table 11 reports these marginal graph-macro gaps; Table 12 gives every fixed, random and classical reference. The random-proposal estimate first averages five frozen seeds (1101–1105) within each condition, then conditions within graph. Its 105 trajectories are not 105 independent graphs.

Table 11: Supplementary model outcomes: graph-macro relative tour gap (%), lower is better, conditional on completed tours. A is reused full-action inference; B offers anonymous proposals; C adds their rule semantics. All scheduled cases remain in coverage. Jev B is missing one condition.

| Model      | Arm | Mean gap (%) | Complete |
|------------|-----|--------------|----------|
| Jev        | A   | 66.01        | 21/21    |
| Jev        | B   | 31.12        | 20/21    |
| Jev        | C   | 27.80        | 21/21    |
| Qwen3.5-4B | A   | 307.11       | 21/21    |
| Qwen3.5-4B | B   | 43.70        | 21/21    |
| Qwen3.5-4B | C   | 36.36        | 21/21    |
| Qwen3.5-9B | A   | 381.82       | 21/21    |
| Qwen3.5-9B | B   | 39.09        | 21/21    |
| Qwen3.5-9B | C   | 26.76        | 21/21    |

Proposal restriction gives much smaller gaps than A, but even uniform random selection among distinct proposals has a 43.18% gap, below all three A model means. Large A-to-B/C improvements therefore cannot be credited entirely to model decisions: the externally supplied heuristics already provide substantial solution quality. Every model arm remains worse on graph-macro gap than the fixed return-reserve rule (16.01%) and NN + 2-opt (5.93%). The minimum-edge selector is exactly

Table 12: All supplementary references, using the same graph-macro gap. Nearest, shortest-edge proposal selection and NN produce identical objectives and represent the same selection rule, not three new algorithms. The random row averages five prespecified seeds per condition. Solve is graph-macro mean local solve time in seconds; wrapper costs differ even for equivalent selectors. None of these references calls a model.

| Reference                              | Gap (%) | Complete | Solve (s) |
|----------------------------------------|---------|----------|-----------|
| Nearest rule                           | 27.65   | 21/21    | 0.069     |
| One-step lookahead                     | 56.86   | 21/21    | 0.070     |
| Return reserve                         | 16.01   | 21/21    | 0.067     |
| Isolation priority                     | 32.16   | 21/21    | 0.068     |
| Uniform distinct proposal (five seeds) | 43.18   | 105/105  | 0.072     |
| Shortest-edge proposal selector (= NN) | 27.65   | 21/21    | 0.072     |
| Nearest neighbor (= nearest rule)      | 27.65   | 21/21    | 0.032     |
| NN + 2-opt                             | 5.93    | 21/21    | 0.089     |

NN because a nearest-city proposal is always present. Reporting this equivalence does not introduce an additional algorithm. The 252 recorded baseline trajectories include five random seeds and these equivalent implementations, not 252 independent graphs.

The primary contrast is paired  $C - B$ . For each graph, compute differences on jointly completed conditions, then average the seven graph-level differences. Jev’s estimate uses twenty pairs across seven graphs and is  $-3.22$  percentage points, not the  $-3.32$  obtained by subtracting the two marginal means. Qwen3.5-4B and Qwen3.5-9B have paired changes of  $-7.34$  and  $-12.32$  points. Directions improve on five, seven and six graphs, respectively (Figure 4). All prespecified contrasts, including the larger A-to-B/C changes, appear in Table 13. They describe these seven reused graphs and correlated conditions, not significance or population effect estimates.

Table 13: All prespecified paired contrasts, in percentage points (pp). Each estimate gives equal weight to the seven graphs after pairing completed conditions within graph. Negative is better for the first arm. Improved-graph counts are descriptive; missing pairs are not imputed.

| Model      | Contrast | Gap change (pp) | Improved graphs | Paired conditions |
|------------|----------|-----------------|-----------------|-------------------|
| Jev        | $C - B$  | -3.22           | 5/7             | 20/21             |
| Jev        | $B - A$  | -33.05          | 7/7             | 20/21             |
| Jev        | $C - A$  | -38.22          | 7/7             | 21/21             |
| Qwen3.5-4B | $C - B$  | -7.34           | 7/7             | 21/21             |
| Qwen3.5-4B | $B - A$  | -263.42         | 7/7             | 21/21             |
| Qwen3.5-4B | $C - A$  | -270.76         | 7/7             | 21/21             |
| Qwen3.5-9B | $C - B$  | -12.32          | 6/7             | 21/21             |
| Qwen3.5-9B | $B - A$  | -342.74         | 7/7             | 21/21             |
| Qwen3.5-9B | $C - A$  | -355.06         | 7/7             | 21/21             |

### H.3 Calls, timing and interpretation

There are 9,451 new model calls and 8,138 forced singleton-proposal transitions (Table 14). The failed episode leaves 195 scheduled decision positions unreachable. Together these exhaust the 17,784 potential B/C decision positions, excluding the forced last city and return. Depending on model and arm, approximately 44–48% of observed proposal steps require no model call. This is part of the abstraction, not evidence of a faster inference architecture.

### Paired tour-gap changes with named append proposals

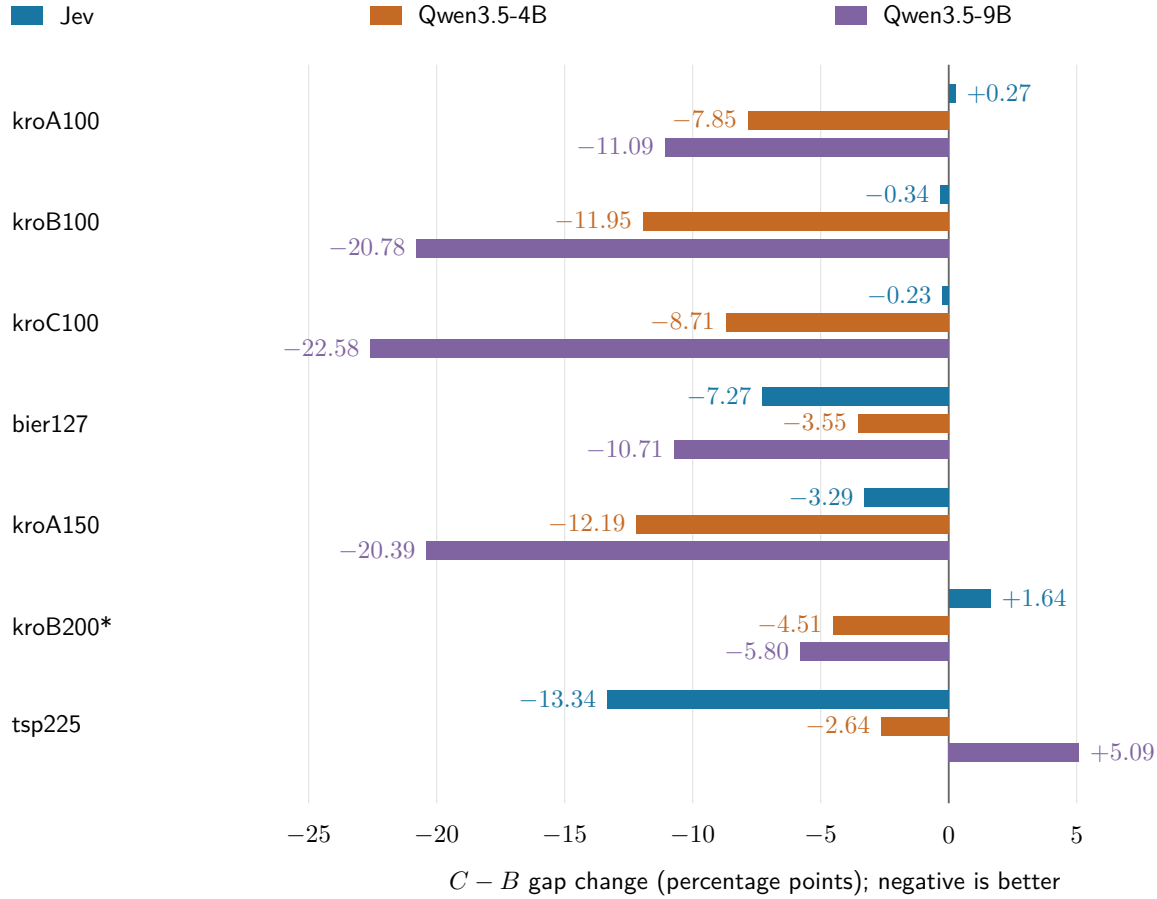


Figure 4: Per-graph paired change from anonymous B to named C proposals. Each bar averages matched completed conditions on one original graph; negative values mean smaller tour gaps. The starred **kroB200** row has two Jev pairs because B condition 1 fails; its Qwen comparisons and all other rows use three pairs. These are observed changes, not confidence intervals.

Table 14: Complete call accounting, including the failed call. A calls are reused, not new inference. Forced proposals count only singleton candidate groups, excluding the deterministic final city and return. Unreached positions belong to the one failed episode.

| Model      | Arm | Model calls | Forced proposals | Unreached positions |
|------------|-----|-------------|------------------|---------------------|
| Jev        | A   | 2964        | 0                | 0                   |
| Jev        | B   | 1475        | 1294             | 195                 |
| Jev        | C   | 1557        | 1407             | 0                   |
| Qwen3.5-4B | A   | 2964        | 0                | 0                   |
| Qwen3.5-4B | B   | 1596        | 1368             | 0                   |
| Qwen3.5-4B | C   | 1646        | 1318             | 0                   |
| Qwen3.5-9B | A   | 2964        | 0                | 0                   |
| Qwen3.5-9B | B   | 1629        | 1335             | 0                   |
| Qwen3.5-9B | C   | 1548        | 1416             | 0                   |

Table 15: Graph-macro mean timing components in seconds, over all scheduled conditions, including partial execution of the failed Jev B episode. Calls measure prediction-plus-adaptor round trips, not provider-internal time. A’s historical wall time includes replay; B/C solve wall excludes independent replay. Dashes mark unavailable or inapplicable measurements, not zero.

| Model / arm  | Setup | Proposals | Calls   | Solve  | Replay | Legacy A wall |
|--------------|-------|-----------|---------|--------|--------|---------------|
| Jev A        | –     | –         | 21.528  | –      | 0.871  | 23.692        |
| Jev B        | 0.016 | 0.113     | 8.939   | 9.149  | 0.243  | –             |
| Jev C        | 0.016 | 0.125     | 8.699   | 8.928  | 0.279  | –             |
| Qwen3.5-4B A | –     | –         | 142.967 | –      | 0.860  | 145.460       |
| Qwen3.5-4B B | 0.016 | 0.126     | 49.443  | 49.675 | 0.280  | –             |
| Qwen3.5-4B C | 0.016 | 0.130     | 51.415  | 51.657 | 0.277  | –             |
| Qwen3.5-9B A | –     | –         | 215.490 | –      | 0.856  | 218.075       |
| Qwen3.5-9B B | 0.016 | 0.129     | 79.925  | 80.165 | 0.270  | –             |
| Qwen3.5-9B C | 0.016 | 0.126     | 77.767  | 78.004 | 0.274  | –             |

B/C solve wall includes setup, candidate computation, model calls, response handling, durable logging and terminal scoring, but excludes independent post-run replay. Setup, proposal computation, model-call time and replay are recorded separately; the components are not additional times to add on top of the inclusive solve clock. A’s earlier audited episode wall time includes replay, and its solve-only wall time is unavailable. The separately measured current replay cost cannot be subtracted from that historical wall time to reconstruct it. Table 15 uses means, unlike the primary panel’s medians; the partial failed episode is execution accounting, not a solved-tour timing. Different call counts, hardware/network paths and concurrent service further prevent a clean A-versus-B/C speedup claim.

The results support a useful but limited diagnostic: proposal-based protocols yield better tours than these models’ full-action runs, and named arm C has a smaller paired mean gap than anonymous B for all three configurations. They do not establish an independently learned planner or solver superiority. The best fixed rule remains better than every model arm; it is an executable prespecified baseline, not an oracle that chooses a rule after seeing tour outcomes. The full graph is retained, yet the action space and forced-transition rate change substantially. A-to-B/C also changes the feature/option representation, not only the number of available actions. Histories, candidate pools and prompts then diverge, so the improvement cannot be assigned solely to semantic understanding. This appendix neither retunes the primary benchmark nor extends its conclusions to other model budgets or graph families.

## I Extended related work

**Graph reasoning benchmarks.** NLGraph evaluates language models on graph problems expressed in natural language, while GraphArena extends graph-computation evaluation to real-world-derived graphs and distinguishes optimal, feasible but suboptimal, infeasible, and missing solutions [24, 28]. These benchmarks establish graph computation and solution verification as existing evaluation directions. GraphWiz uses the GraphInstruct instruction-tuning corpus to train models that produce explicit reasoning paths for graph problems [3]. GraphLLM integrates graph learning models with language models, illustrating that graph representation and model design can change together [2]. CLRS provides a complementary algorithmic-learning perspective spanning graph algorithms and other classical procedures [27]. Our study examines exact decisions and bounded-choice construction under interface and verification controls, rather than proposing a

graph-trained model or a general algorithmic-learning benchmark.

**Learned constructive optimization.** Sequential neural decisions for combinatorial optimization predate the systems evaluated here. Khalil et al. combine graph embeddings and reinforcement learning to learn greedy constructive policies for problems including MaxCut and TSP; Kool et al. train attention-based routing policies using the REINFORCE policy-gradient algorithm with a greedy rollout baseline [12, 13]. Our tasks share the incremental-decision setting but evaluate general-purpose decision interfaces without claiming to replace specialized learned policies. HeurAgenix studies LLM-guided heuristic development and selection for combinatorial optimization [29]. Selecting a heuristic delegates its internal decisions to an algorithm; GraphDecide’s construction interface instead asks the model to choose each city, assignment or seed directly. The two settings expose different action spaces and levels of algorithmic assistance. The community task uses standard modularity [18], restricted to at most two communities; it is not unrestricted community detection or recovery of ground-truth communities. Likewise, our influence task uses fixed thresholds and deterministic diffusion. It must not be conflated with the classical independently randomized-threshold LT model whose expected spread is submodular [11]. That model’s greedy approximation guarantee is not assumed here.

**Typed decisions and contemporary systems.** TypeSafe’s launch article describes Jev as returning typed probabilistic decisions rather than generated strings [1]. Its model documentation identifies weaknesses in numeric precision and multi-step indirection [26]. These are vendor descriptions, not independent architecture or graph-performance evidence. Two contemporary software projects are relevant [14, 25]. SemIf implements direct option-logit readout from open models and distinguishes reproducing an interface pattern from reproducing Jev’s undisclosed model. WorldTour already evaluates Jev-guided next-city choices for heuristic routing and disclaims global optimality of its large tour. We therefore make no priority claim for typed-choice evaluation or Jev-guided TSP.

**Typed-decision leaderboards.** JevBench evaluates bounded decisions across routing, policy application, answer judgment and related reasoning tasks [5]. Its v1.4.2 ranking combines accuracy-related and calibration axes with deployment-dependent speed and cost; outranking Jev therefore need not mean better decision accuracy or predictive-distribution equivalence. The published label-scoring protocol does not verify complete graph constructions. That label-scoring contract differs from evaluating complete graph objectives and accumulated own-history construction errors, irrespective of whether some private items concern graphs. Leaderboard ranks provide context for candidate selection, not evidence of transfer to our tasks.

**Scope of this benchmark.** In the synthetic suite, matched query pairs, explicit identity checks and exactly scored own-history construction address particular shortcuts and accumulated decision errors. It does not establish immunity to every shortcut or training-data contamination. Same-backbone interface comparisons do not identify a closed service’s architecture; exact performance on tiny instances does not establish large-graph scaling or general solver superiority.

**Predictive reliability.** Returning a numeric distribution is not evidence of calibrated uncertainty. Proper scoring rules evaluate the predictive distribution against outcomes [6]; modern-network calibration studies distinguish classification accuracy from confidence reliability [7]. These motivate separate probability diagnostics rather than treating a typed API or normalized option logits as

calibrated by construction. We do not transfer published calibration findings to the present models without data.

## J Synthetic query accuracies

Table 16 provides the numerical values visualized in Figure 3. All twelve configurations return legal answers on all 800 query cases. Each metric first computes accuracy within each of the five graph sizes and then averages those five values equally. For degree, the  $8n$  cases per size make this macro accuracy differ from the pooled fraction correct. Qwen3.8-27B and Qwen2.5-72B-Instruct have degree macro accuracies of 45.50% and 41.24%, respectively, versus pooled accuracies of 44.50% and 40.50%. Neither aggregation is an across-task score.

Table 16: Synthetic exact-task macro-size accuracy (%). Model identity and decision interface are separate. Grammar denotes greedy legal-choice generation; Qwen3.5 and Qwen3.8 thinking is disabled. The lower block contains separately budgeted GPT reasoning references.

| Model                | Interface         | Adjacency | Degree | Cycle  | Connectivity | Distance | Articulation |
|----------------------|-------------------|-----------|--------|--------|--------------|----------|--------------|
| Jev-1.13.0           | Native            | 100.00    | 61.43  | 66.25  | 85.00        | 58.75    | 65.00        |
| Qwen3.5-4B           | Grammar           | 60.00     | 18.32  | 55.00  | 50.00        | 50.00    | 50.00        |
| Qwen3.5-9B           | Grammar           | 58.75     | 26.99  | 51.25  | 53.75        | 42.50    | 50.00        |
| Qwen3.5-27B          | Grammar           | 98.75     | 47.70  | 63.75  | 80.00        | 52.50    | 47.50        |
| Qwen3.8-27B          | Grammar           | 90.00     | 45.50  | 61.25  | 85.00        | 48.75    | 50.00        |
| Qwen2.5-72B-Instruct | Grammar           | 92.50     | 41.24  | 50.00  | 86.25        | 60.00    | 46.25        |
| Qwen3.5-4B           | Candidate scoring | 77.50     | 23.89  | 51.25  | 51.25        | 52.50    | 50.00        |
| Decider-2B           | Native            | 58.75     | 12.09  | 50.00  | 50.00        | 46.25    | 50.00        |
| Kev-4B               | Native            | 78.75     | 19.88  | 57.50  | 58.75        | 51.25    | 45.00        |
| Laya                 | Native            | 50.00     | 10.87  | 50.00  | 50.00        | 50.00    | 50.00        |
| GPT-5.4              | Default reasoning | 100.00    | 100.00 | 100.00 | 100.00       | 100.00   | 100.00       |
| GPT-6-Astra          | Default reasoning | 100.00    | 100.00 | 100.00 | 100.00       | 100.00   | 100.00       |

## K Synthetic capability profile and interpretation

### K.1 RQ1: Construction needs graded quality and coverage

Within this synthetic bank, mean additive gap measures construction quality. TSP illustrates its discrimination: Jev’s gap is 45.85, versus 100.90 for Qwen3.5-27B and 349.65 for Kev. LT supplies a complementary regime: Qwen3.5-27B has a smaller gap than Jev (0.425 versus 0.600), while Kev and Jev tie at 0.600. Community gaps favor Jev over those two configurations (0.1694 versus 0.2810 and 0.2502). All three complete the same forty episodes per task; comparisons involving incomplete configurations must also account for coverage. Optimum-hit counts can conceal large quality differences or suggest an ordering absent from mean gaps. The added Qwen3.8-27B and Qwen2.5-72B have larger mean gaps than Jev on all four constructions. This is a task-specific comparison of configurations, not evidence of general 72B-model equivalence: generation, latency and training-data exposure are not controlled.

Classical references provide a separate quality threshold. In the original four-configuration challenge, all neural MaxCut gaps exceed the exact uniform-policy expected gap (Jev: 8.150, uniform: 5.325); classical cut search has mean gap 0.450. Nearest-neighbor plus 2-opt has mean TSP gap 0.200 versus Jev’s 45.850. These revisable algorithms are quality references, not equal-budget online competitors (Appendix O).

## K.2 RQ2: Discrimination is task dependent

The benchmark distinguishes several capabilities rather than defining one hardness scale. On the canonical challenge, Jev answers every adjacency query correctly, whereas Kev reaches 78.75%, Decider 58.75%, and Laya 50%. Jev’s degree macro accuracy is 61.43%, versus 19.88% for Kev, and its connectivity accuracy is 85%, versus 58.75%. Conversely, distance accuracy is 58.75% for Jev and 42.5–52.5% for the earlier direct configurations; the added Qwen3.8-27B and Qwen2.5-72B score 48.75% and 60%, respectively. Close scores around the balanced binary baseline do not support a stable fine-grained ranking.

Consecutive graph sizes offer descriptive resolution, not identified capacity thresholds. Jev degree accuracy is 70.3% at eight vertices and 49.0% at twelve; cycle accuracy is 81.25% and 50%, respectively, with nonmonotone intermediate values. Connectivity does not exhibit the same decline, and adjacency remains perfect throughout. The degree task also increases its number of classes with size. These observations cannot be reduced to “Jev fails at twelve vertices.” Each binary size cell contains only sixteen queries in eight pairs; each construction cell has eight episodes. We therefore do not assign empirically fitted reliability bands or claim a statistically identified breakpoint.

## K.3 RQ3: Replication and controls

The prospective 952-instance, topology-excluded confirmation covers the original four configurations, not all twelve. All four again have larger MaxCut gaps than the uniform legal policy; Jev retains perfect adjacency accuracy. Individual rankings do change, so this supports selected qualitative findings rather than every canonical point estimate (Appendix P).

Controls distinguish behavioral weakness from several identifiable artifacts. Full native-score and action replay retains Laya’s constant-yes outcome; its selected option-reversal pilot argues against a simple first-option explanation, not against every possible prompting effect. For the original local baselines, identical aggregate accuracy can conceal complete semantic-choice reversal. At a fixed token prefix, 4B score argmax and masked greedy generation agree on 100/100 states, preventing attribution of the main pipeline’s gains to argmax alone. Jev’s above-chance distance accuracy also coexists with probability losses worse than uniform on the common-valid probability-audit rows. These checks address distinct failure modes; they are not a proof of freedom from all implementation errors (Appendix Q).

## K.4 Supplementary trajectory diagnostic

Exact continuation analysis locates losses of reachable optimality in recorded synthetic trajectories. Across 640 constructions and 4,480 decisions of the original four configurations, the sum of stepwise losses matches the final objective gap in every episode. For TSP, optimality is already lost at the first choice in 3/40 Jev episodes, compared with 22/40 Qwen3.5-4B grammar episodes. Later legal actions can preserve the best remaining completion without restoring the original optimum. This retrospective calculation does not supply oracle actions during inference or identify the cause of a model’s choices (Appendix O.3).

**Interpretation.** The synthetic suite contains positive controls, partially successful structural decisions, and construction failures with measurable quality differences. This is enough to characterize task-dependent strengths and weaknesses within the tested protocols. It does not locate the onset of every failure or establish size generalization. Fresh-graph confirmation of the added readouts and controlled variation in graph structure remain separate tests.

## L Supplementary synthetic construction quality

Every configuration completes all 960 synthetic cases except Laya and GPT-5.4. Laya rejects nine LT inputs at its lossless context/head limits, leaving 31/40 LT episodes. GPT-5.4’s nine construction timeouts leave 36/40 cut, 39/40 TSP, 40/40 LT and 36/40 modularity episodes. It attains zero conditional gap on 151/160 feasible constructions, whereas GPT-6-Astra attains zero gap on all 160. Table 18 retains the full scheduled accounting.

These additive gaps concern the small synthetic bank, not the public relative-gap panel. Jev’s mean TSP gap is 45.85 versus 100.90 for Qwen3.5-27B and 349.65 for Kev (Table 17). The LT ordering reverses: Jev and Qwen3.5-27B have gaps 0.600 and 0.425. Kev ties Jev at 0.600 despite a larger optimum-hit count. Qwen3.5-27B has a slightly smaller cut gap (8.025 versus 8.150), whereas Jev has a smaller modularity gap (0.1694 versus 0.2810). Jev also has smaller gaps on all four tasks than Qwen3.8-27B and Qwen2.5-72B; their TSP gaps are 183.70 and 152.40. These findings concern the tested configurations, not parameter equivalence or a monotone model-size ordering.

Table 17: Synthetic construction quality: mean additive objective gap over feasible completions; lower is better. All cells use forty episodes except Laya LT (31), GPT-5.4 cut (36), TSP (39) and modularity (36). Units differ across columns; differing coverage is not a matched-subset comparison. The GPT block uses separate reasoning budgets.

| Model                | Interface         | MaxCut | TSP    | Deterministic LT | Modularity |
|----------------------|-------------------|--------|--------|------------------|------------|
| Jev-1.13.0           | Native            | 8.150  | 45.85  | 0.600            | 0.1694     |
| Qwen3.5-4B           | Grammar           | 11.125 | 338.15 | 2.625            | 0.2900     |
| Qwen3.5-9B           | Grammar           | 11.050 | 306.45 | 3.725            | 0.3203     |
| Qwen3.5-27B          | Grammar           | 8.025  | 100.90 | 0.425            | 0.2810     |
| Qwen3.8-27B          | Grammar           | 11.150 | 183.70 | 2.675            | 0.3012     |
| Qwen2.5-72B-Instruct | Grammar           | 9.775  | 152.40 | 1.950            | 0.2645     |
| Qwen3.5-4B           | Candidate scoring | 9.200  | 351.00 | 4.125            | 0.2810     |
| Decider-2B           | Native            | 11.125 | 355.70 | 4.100            | 0.2810     |
| Kev-4B               | Native            | 11.075 | 349.65 | 0.600            | 0.2502     |
| Laya                 | Native            | 15.250 | 344.75 | 2.871            | 0.3039     |
| GPT-5.4              | Default reasoning | 0.000  | 0.00   | 0.000            | 0.0000     |
| GPT-6-Astra          | Default reasoning | 0.000  | 0.00   | 0.000            | 0.0000     |

Classical references put these differences in perspective. Jev’s MaxCut gap of 8.150 exceeds the exact uniform-policy expected gap of 5.325; classical cut search achieves 0.450. Nearest-neighbor tours refined by 2-opt have TSP gap 0.200 versus Jev’s 45.850. These algorithms may revise choices and are solution-quality references, not equal-budget online policies. An advantage over another neural configuration need not mean competitive optimization quality.

## M Synthetic configuration coverage

Table 18: Synthetic diagnostic coverage. Exact feasibility means a valid answer, not a correct answer; construction feasibility requires a complete legal trajectory. Input rejects and execution failures are episode counts. All twelve configurations retain 960 scheduled cases. The lower block contains separately budgeted GPT reasoning references.

| Model                | Interface         | Exact feasible | Construct feasible | Input rejects | Exec. failures | Unattempted |
|----------------------|-------------------|----------------|--------------------|---------------|----------------|-------------|
| Jev-1.13.0           | Native            | 800/800        | 160/160            | 0             | 0              | 0           |
| Qwen3.5-4B           | Grammar           | 800/800        | 160/160            | 0             | 0              | 0           |
| Qwen3.5-9B           | Grammar           | 800/800        | 160/160            | 0             | 0              | 0           |
| Qwen3.5-27B          | Grammar           | 800/800        | 160/160            | 0             | 0              | 0           |
| Qwen3.8-27B          | Grammar           | 800/800        | 160/160            | 0             | 0              | 0           |
| Qwen2.5-72B-Instruct | Grammar           | 800/800        | 160/160            | 0             | 0              | 0           |
| Qwen3.5-4B           | Candidate scoring | 800/800        | 160/160            | 0             | 0              | 0           |
| Decider-2B           | Native            | 800/800        | 160/160            | 0             | 0              | 0           |
| Kev-4B               | Native            | 800/800        | 160/160            | 0             | 0              | 0           |
| Laya                 | Native            | 800/800        | 151/160            | 9             | 0              | 0           |
| GPT-5.4              | Default reasoning | 800/800        | 151/160            | 0             | 9              | 0           |
| GPT-6-Astra          | Default reasoning | 800/800        | 160/160            | 0             | 0              | 0           |

## N Supplementary construction optimum-hit counts

The synthetic construction comparison uses mean objective gaps and feasible coverage (Table 17). Counts below are retained for continuity with the frozen experiment outputs; they are not the primary quality metric.

Table 19: Supplementary complete canonical optimum hits over all forty scheduled episodes per task. All tied optima are accepted. Laya’s nine LT input rejects and GPT-5.4’s nine timeouts remain in denominators.

| Model                | Interface         | MaxCut | TSP   | Deterministic LT | Modularity |
|----------------------|-------------------|--------|-------|------------------|------------|
| Jev-1.13.0           | Native            | 0/40   | 3/40  | 23/40            | 7/40       |
| Qwen3.5-4B           | Grammar           | 0/40   | 0/40  | 13/40            | 0/40       |
| Qwen3.5-9B           | Grammar           | 0/40   | 0/40  | 5/40             | 0/40       |
| Qwen3.5-27B          | Grammar           | 0/40   | 1/40  | 29/40            | 0/40       |
| Qwen3.8-27B          | Grammar           | 0/40   | 0/40  | 10/40            | 0/40       |
| Qwen2.5-72B-Instruct | Grammar           | 0/40   | 1/40  | 5/40             | 0/40       |
| Qwen3.5-4B           | Candidate scoring | 0/40   | 0/40  | 3/40             | 0/40       |
| Decider-2B           | Native            | 0/40   | 0/40  | 1/40             | 0/40       |
| Kev-4B               | Native            | 0/40   | 0/40  | 26/40            | 1/40       |
| Laya                 | Native            | 0/40   | 0/40  | 6/40             | 0/40       |
| GPT-5.4              | Default reasoning | 36/40  | 39/40 | 40/40            | 36/40      |
| GPT-6-Astra          | Default reasoning | 40/40  | 40/40 | 40/40            | 40/40      |

## O Original synthetic-challenge results

### O.1 Primitive correctness is task dependent

Jev obtains 100% adjacency accuracy but 58.75% distance and 65% articulation accuracy (Table 20). Its degree macro accuracy is 61.43%, whereas pooled accuracy is 241/400, or 60.25%. Reporting only the direct-lookup ceiling would obscure these differences.

Table 20: Exact-task macro-size accuracy (%). All four methods complete all 960 challenge instances. Each binary task has eighty cases; degree has 400. These are bounded configurations, not unrestricted model-capacity rankings.

| Exact task (macro %) | Jev-1.13.0<br>(Native) | Qwen3.5-4B<br>(Grammar) | Qwen3.5-9B<br>(Grammar) | Qwen3.5-4B<br>(Candidate scoring) |
|----------------------|------------------------|-------------------------|-------------------------|-----------------------------------|
| Adjacency            | 100.00                 | 60.00                   | 58.75                   | 77.50                             |
| Degree               | 61.43                  | 18.32                   | 26.99                   | 23.89                             |
| Cycle                | 66.25                  | 55.00                   | 51.25                   | 51.25                             |
| Connectivity         | 85.00                  | 50.00                   | 53.75                   | 51.25                             |
| Distance             | 58.75                  | 50.00                   | 42.50                   | 52.50                             |
| Articulation         | 65.00                  | 50.00                   | 50.00                   | 50.00                             |

Table 21: Optimal solutions over forty scheduled episodes per construction task. Classical references use separate revisable full-solution algorithms. All model episodes in this table are feasible.

| Task             | Jev-1.13.0<br>(Native) | Qwen3.5-4B<br>(Grammar) | Qwen3.5-9B<br>(Grammar) | Qwen3.5-4B<br>(Candidate scoring) | Classical |
|------------------|------------------------|-------------------------|-------------------------|-----------------------------------|-----------|
| MaxCut           | 0/40                   | 0/40                    | 0/40                    | 0/40                              | 29/40     |
| TSP              | 3/40                   | 0/40                    | 0/40                    | 0/40                              | 37/40     |
| Deterministic LT | 23/40                  | 13/40                   | 5/40                    | 3/40                              | 35/40     |
| Modularity       | 7/40                   | 0/40                    | 0/40                    | 0/40                              | 35/40     |

Candidate scoring improves 4B adjacency from 60% to 77.5% but does not uniformly improve other tasks. Nor does the bounded 9B configuration uniformly improve on 4B. Binary chance-level results are accompanied by prediction counts and per-label/per-degree recall, since a constant answer can achieve 50% on balanced pairs without discriminating either case. The size curves (Figure 5) are descriptive; topology, serialization length and action cardinality also vary.

## O.2 Legal construction does not imply competitive optimization

All four original bounded configurations miss every MaxCut optimum; their mean gaps also exceed the uniform-policy expected gap (Table 22). Classical cut search reaches 29/40 optima. For TSP, Jev reaches 3/40 optima versus 37/40 for nearest-neighbor plus 2-opt. Thus neither legal output nor relative improvement over a weak neural configuration implies a useful solver. On LT, 4B scoring attains only 3/40 optima versus 13/40 for 4B generation; the same readout that improves adjacency can harm sequential performance.

The suspicious cut and modularity floors triggered independent NetworkX objective recomputation, with zero mismatches. The 4B grammar configuration assigns every remaining vertex to B in all forty cuts; 9B does so in 39/40, scoring in 20/40, and Jev in 13/40. The scoring method chooses all-A on every modularity episode, legally obtaining  $Q = 0$ . These are observed policies, not proofs of an internal failure mechanism or an exhaustive absence of bugs.

## O.3 Where actual trajectories lose optimality

This supplementary analysis uses exact optima on the small synthetic graphs; it is not performed on the public-instance construction panel. Let  $s_t$  contain the model’s choices through step  $t$ , and let

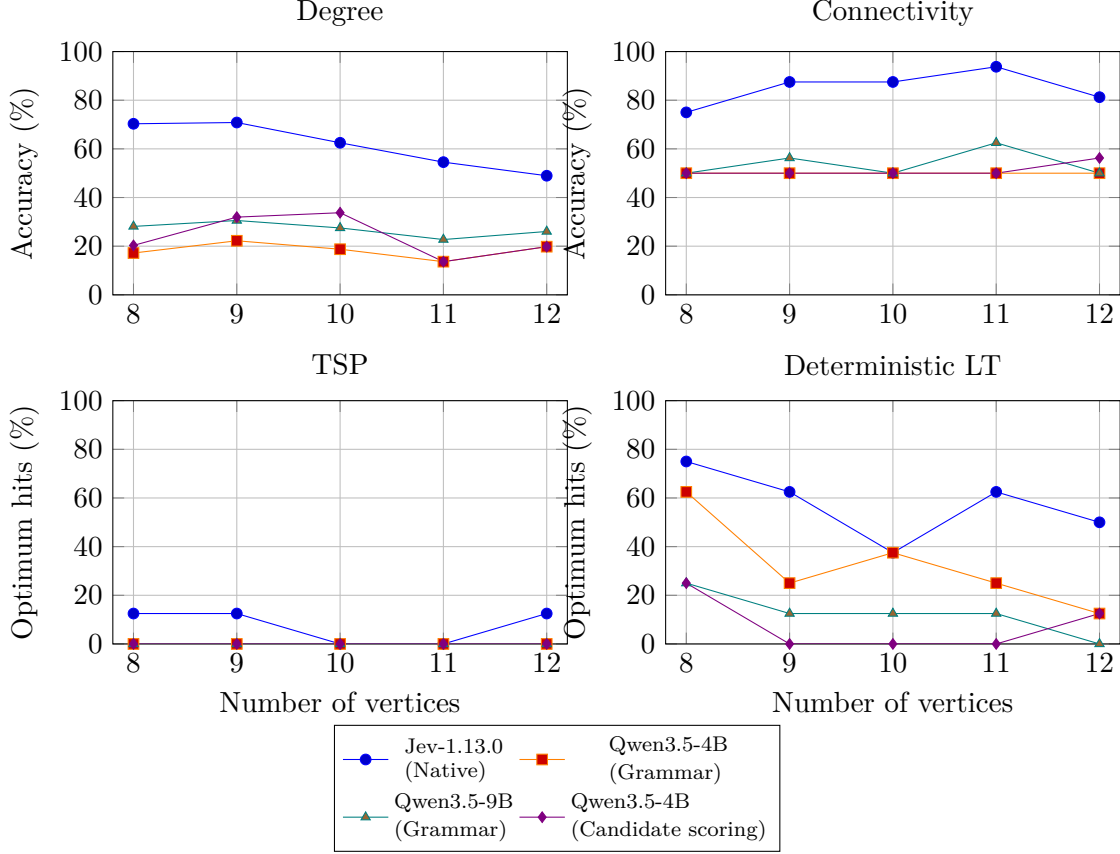


Figure 5: Selected size curves on the structural challenge. Exact-task accuracy and construction optimum-hit rate are different metrics. Each construction point has eight episodes; curves do not imply monotone or size-causal difficulty. Full task/size counts are in the artifact.

$V(s_t)$  be the exact best final objective among its legal completions. Define

$$r_t = \begin{cases} V(s_{t-1}) - V(s_t), & \text{maximization,} \\ V(s_t) - V(s_{t-1}), & \text{minimization.} \end{cases} \quad r_t \geq 0, \quad \sum_{t=1}^T r_t = \text{gap}. \quad (1)$$

Completion sets shrink as choices accumulate. For a complete trajectory,  $V(s_0) = f^*$  and  $V(s_T) = f$ , giving the telescoping identity. A zero-loss action preserves the best reachable completion, not a global optimum excluded earlier. This accounting allows tied optima and does not identify a causal mechanism of model behavior.

We compute this diagnostic *after inference* from actual histories, using exact prefix-conditioned continuations. Across 640 complete construction episodes and 4,480 decisions, every decomposition equals the independently replayed final gap within  $10^{-12}$ . Table 23 shows how often optimality is lost immediately. For MaxCut, only 7–8 of forty episodes lose optimality at the first action despite every complete neural solution being suboptimal. In TSP, the first action already loses optimality in 22/40 grammar-4B and 29/40 scoring episodes, versus 3/40 for Jev. No oracle continuation value was returned to the model.

Table 22: Mean additive objective gap, lower is better. Uniform is the exact expected gap of uniform legal decisions, not a sampled solver. Columns are comparable within a row only; the four objectives have different units.

| Task             | Jev-1.13.0<br>(Native) | Qwen3.5-4B<br>(Grammar) | Qwen3.5-9B<br>(Grammar) | Qwen3.5-4B<br>(Candidate scoring) | Classical | Uniform |
|------------------|------------------------|-------------------------|-------------------------|-----------------------------------|-----------|---------|
| MaxCut           | 8.150                  | 11.125                  | 11.050                  | 9.200                             | 0.450     | 5.325   |
| TSP              | 45.850                 | 338.150                 | 306.450                 | 351.000                           | 0.200     | 331.288 |
| Deterministic LT | 0.600                  | 2.625                   | 3.725                   | 4.125                             | 0.175     | 4.182   |
| Modularity       | 0.169                  | 0.290                   | 0.320                   | 0.281                             | 0.007     | 0.338   |

Table 23: Episodes losing every globally optimal completion at the first decision, over all forty feasible episodes. Later errors remain represented in the full per-step regret artifact. This is a retrospective diagnostic, not a new inference experiment.

| Task             | Jev-1.13.0<br>(Native) | Qwen3.5-4B<br>(Grammar) | Qwen3.5-9B<br>(Grammar) | Qwen3.5-4B<br>(Candidate scoring) |
|------------------|------------------------|-------------------------|-------------------------|-----------------------------------|
| MaxCut           | 8/40                   | 8/40                    | 8/40                    | 7/40                              |
| TSP              | 3/40                   | 22/40                   | 21/40                   | 29/40                             |
| Deterministic LT | 8/40                   | 12/40                   | 24/40                   | 26/40                             |
| Modularity       | 12/40                  | 10/40                   | 23/40                   | 23/40                             |

#### O.4 Latency and native conformance are separate outcomes

Across development and challenge, the four-method matrix makes 10,352 actual calls for 5,264 scheduled instance evaluations. Jev retains 25 strict native observations: 24 probability-sum checks and one choice/argmax check. All supplied actions are eligible under the separately frozen action protocol. We neither normalize probabilities nor infer replacement choices. The unchanged strict production parser and its historical results remain available as a different protocol.

The challenge scoring method evaluates 8,880 candidates in 1,920 prefixes, with 1,920 unused generated tokens. Median request latency is 64.41 ms, compared with Jev’s 70.01 ms (Table 24). This is not evidence that one architecture is inherently faster. Vendor-reported tokens are preserved but are not interchangeable autoregressive steps, floating-point operations (FLOPs), energy, or monetary cost.

Table 24: Observed client-path request latency on the challenge. Network, rendering, cache and hosting differ; this is not an architecture-isolated speed comparison. Here p95 denotes the 95th percentile.

| Method                         | Median (ms) | p95 (ms) |
|--------------------------------|-------------|----------|
| Jev-1.13.0 (Native)            | 70.01       | 115.55   |
| Qwen3.5-4B (Grammar)           | 94.24       | 146.18   |
| Qwen3.5-9B (Grammar)           | 148.64      | 235.12   |
| Qwen3.5-4B (Candidate scoring) | 64.41       | 139.83   |

## P Prospective new-seed confirmation

After fixing the common-interface protocol and four configurations, we registered a separate confirmation before generating or evaluating its graphs. Seed 20261012, sizes 8–12, eight repetitions and the same 128-proposal cap were fixed. The prior pool includes the earlier exclusion pool, both inspected development banks, the original challenge, and forty executed TSP confirmation metrics: 2,052 labeled normalized graph records after deduplication. These are not 2,052 isomorphism classes. Independent graph reconstruction and 1,421 exact VF2 comparisons after necessary invariant screening found zero prior overlaps.

The generator fills 952/960 requested instances. Four cycle pairs at  $n = 8$  exhaust the finite proposal budget, leaving eight explicitly missing cases; cycle has 72 instances and every other task reaches its quota. The seed is not retried and no unmatched fallback is introduced. This limitation affects coverage and precision, not just implementation bookkeeping: only four cycle-pair clusters remain at  $n = 8$ . The repeated task/size quotas are not changed after observing any model answer.

All four methods complete all 952 realized cases, using 1,912 calls each: 3,808 instance evaluations and 7,648 actual calls. The eight Jev probability-sum observations remain conformance diagnostics; no action execution fails. Sealed attempts, scores and requests replay successfully, and initial model-visible requests agree across all methods. The same diagnostic exporter checks another 640 construction trajectories and 4,480 decisions, again with zero continuation-loss telescoping mismatches.

The confirmation preserves the principal qualitative observations without reproducing every numerical ranking. Jev retains its adjacency ceiling; 4B scoring again improves adjacency over 4B grammar (85% versus 63.75%). All methods again miss every MaxCut optimum and have larger expected gap than the uniform legal policy. TSP optimum hits are 2/40 for Jev, zero for the other neural configurations, and 40/40 for the classical reference. Such a classical ceiling is evidence that these particular small TSP instances are controls for neural interfaces, not difficult classical optimization benchmarks. On LT, scoring reaches 2/40 optima versus 6/40 for 4B grammar, retaining the task-dependent tradeoff.

Conversely, 9B distance accuracy changes from 42.5% on the inspected bank to 57.5% here; an individual ranking from one small sample is not a universal capacity claim. Jev-minus-4B-grammar articulation accuracy is 8.75 percentage points with pointwise interval [0,17.5], not a separately established family-wise significant improvement. Full paired intervals appear in the appendix. This confirmation addresses reuse of inspected graphs, but not unseen generator families, larger sizes, pretraining contamination or closed-model service drift.

Table 25: Prospective confirmation exact macro-size accuracy (%). Cycle has 72 cases, degree 400, and the other binary tasks eighty each. The macro estimator weights the five sizes equally, not by their case counts.

| Exact task (macro %) | Jev-1.13.0<br>(Native) | Qwen3.5-4B<br>(Grammar) | Qwen3.5-9B<br>(Grammar) | Qwen3.5-4B<br>(Candidate scoring) |
|----------------------|------------------------|-------------------------|-------------------------|-----------------------------------|
| Adjacency            | 100.00                 | 63.75                   | 58.75                   | 85.00                             |
| Degree               | 64.15                  | 17.38                   | 25.73                   | 20.26                             |
| Cycle                | 71.25                  | 51.25                   | 53.75                   | 50.00                             |
| Connectivity         | 87.50                  | 50.00                   | 55.00                   | 53.75                             |
| Distance             | 58.75                  | 50.00                   | 57.50                   | 50.00                             |
| Articulation         | 58.75                  | 50.00                   | 50.00                   | 50.00                             |

Table 26: Prospective confirmation optimal solutions over forty scheduled episodes per construction. All episodes are feasible; classical algorithms are independently recomputed on these new graphs.

| Task             | Jev-1.13.0<br>(Native) | Qwen3.5-4B<br>(Grammar) | Qwen3.5-9B<br>(Grammar) | Qwen3.5-4B<br>(Candidate scoring) | Classical |
|------------------|------------------------|-------------------------|-------------------------|-----------------------------------|-----------|
| MaxCut           | 0/40                   | 0/40                    | 0/40                    | 0/40                              | 20/40     |
| TSP              | 2/40                   | 0/40                    | 0/40                    | 0/40                              | 40/40     |
| Deterministic LT | 21/40                  | 6/40                    | 2/40                    | 2/40                              | 34/40     |
| Modularity       | 2/40                   | 0/40                    | 0/40                    | 0/40                              | 34/40     |

Table 27: Prospective confirmation mean additive gaps. Uniform columns are exact policy expectations. No graph, task or method is selected for inclusion based on its observed score.

| Task             | Jev-1.13.0<br>(Native) | Qwen3.5-4B<br>(Grammar) | Qwen3.5-9B<br>(Grammar) | Qwen3.5-4B<br>(Candidate scoring) | Classical | Uniform |
|------------------|------------------------|-------------------------|-------------------------|-----------------------------------|-----------|---------|
| MaxCut           | 9.000                  | 11.575                  | 11.375                  | 9.825                             | 0.725     | 5.475   |
| TSP              | 45.300                 | 315.950                 | 298.250                 | 345.100                           | 0.000     | 334.111 |
| Deterministic LT | 0.825                  | 2.600                   | 3.575                   | 3.425                             | 0.250     | 3.677   |
| Modularity       | 0.209                  | 0.290                   | 0.294                   | 0.269                             | 0.005     | 0.326   |

## Q Interface controls and stronger-model context

**Graph-preserving presentation changes.** We preregistered 120 exact-task cases from the confirmation bank: four per task and size. Binary tasks use two complete query/construction pairs; degree uses four prespecified degree strata, not every class. Original, identical-repeat, node-relabeling, edge-shuffling and option-reversal arms produce 2,400 decisions across the four methods. Independent NetworkX checks preserve every transformed label. The relabeling consistently transforms queries and edge endpoints without also shuffling edge positions. Two edge permutations leave the actual serialized input unchanged; both are retained and identified rather than replaced.

All actions are valid. Four Jev strict-native observations are retained separately. Adjacency remains robust for Jev on all twenty selected cases, but its degree correctness falls from 15/20 in the original arm to 11/20 after relabeling; only 9/20 are correct across all five arms (Table 28). An identical repeat already changes two Jev degree choices, so individual flips cannot all be causally attributed to the intervention.

Conversely, 4B grammar connectivity chooses the same answer across all five arms on every case, yet is consistently wrong on ten of twenty cases. For 9B articulation, original and reversed-option accuracy are both 10/20, yet reversing options flips all twenty semantic choices; identical repeats, relabeling and edge shuffling flip none. No case is correct across all five arms. Marginal accuracy can therefore hide severe semantic-choice instability; invariant wrong output is not successful robustness. This is a finite descriptive panel with dependent query pairs, one repeat, and fixed arm ordering, not a population guarantee or a comparison over 600 independent graphs per method.

**Isolating the readout at a fixed token prefix.** The main 4B comparison changes rendering, label coding and decoding, not only the mathematical selection rule. To test the latter directly, we registered 100 initial states: two opaque-ID-first cases per task and size from the completed confirmation bank. Three rotating-order arms use exactly the same prompt token IDs and candidate tokens: requested candidate scores, an identical repeat, and candidate-masked greedy one-token generation. The greedy action is mapped from its actually emitted token; it is not substituted by a post-hoc argmax. The generated token is ignored only in the score arms, and all generated tokens are counted.

All 300 calls succeed. Score argmax and greedy generation agree on 100/100 choices, as do the score repeats. The largest conditional-probability difference is exactly zero in these observed responses. Payload replay checks identical prefixes, the mask, candidate mapping and token accounting. This is consistent with the elementary equivalence of greedy selection and argmax on the same logits; it is not a universal backend equivalence or a successful CPU SemIf reproduction. In particular, the main adjacency gain must be described as a *decision-pipeline* effect, not an isolated benefit of using argmax. Which remaining rendering or coding component causes that gain is not identified by this control. Construction states here are initial states only, not new rollout scores.

**Teacher-forced controls are not rollout scores.** On 24 development graphs, 48 informative optimum-prefix states were tested with original, repeated, reversed-option and explicit-description arms: 384 real decisions across Jev and constrained 9B. Relative to the original, 9B changed 15/48 valid choices under option reversal and 0/48 on identical repeats; Jev changed 5/46 and 2/46. Repetition variability prevents attributing every changed choice to the treatment. Both methods solve all six late MaxCut original-arm states, despite poor free-running cut performance. This illustrates why favorable oracle-prefix decisions cannot substitute for own-history episodes.

Table 28: Cases correct in all five presentation arms, over twenty selected cases per task. Full arm accuracies and paired choice-flip counts appear in the appendix. The degree panel uses four prespecified degree strata per size.

| Task         | Jev-1.13.0<br>(Native) | Qwen3.5-4B<br>(Grammar) | Qwen3.5-9B<br>(Grammar) | Qwen3.5-4B<br>(Candidate scoring) |
|--------------|------------------------|-------------------------|-------------------------|-----------------------------------|
| Adjacency    | 20/20                  | 10/20                   | 11/20                   | 8/20                              |
| Degree       | 9/20                   | 1/20                    | 3/20                    | 3/20                              |
| Cycle        | 9/20                   | 4/20                    | 10/20                   | 10/20                             |
| Connectivity | 13/20                  | 10/20                   | 10/20                   | 8/20                              |
| Distance     | 10/20                  | 10/20                   | 5/20                    | 0/20                              |
| Articulation | 9/20                   | 1/20                    | 0/20                    | 0/20                              |

**Prospective probability diagnostics.** Earlier canonical Jev logs do not retain complete probability vectors, so their calibration cannot be reconstructed from choices or sum diagnostics. We instead register new calls on all 792 exact cases of the confirmation bank, for Jev action and 4B candidate scores: 1,584 observations. Raw vector entries are neither normalized nor clipped. Probability scores require action eligibility, exact option keys, finite entries in  $[0, 1]$  and mass within the frozen  $10^{-6}$  tolerance. Omitted vectors and all-scheduled action accuracy remain separate outcomes.

For accepted vectors, summed multiclass Brier loss is  $\sum_k (p_k - \mathbf{1}[k = y])^2$  and logarithmic loss is  $-\log p_y$ . A zero true-label probability has positive-infinite loss; it is not replaced by a small epsilon or silently dropped from an average. Binary uniform Brier is therefore 0.5 under this summed convention, not 0.25. We compare both models and the uniform reference only on the common-valid intersection, alongside each model’s own retained-vector count. Five fixed reliability bins use the probability of the *actually supplied choice*, not an inferred replacement argmax or an assumed interpretation of a vendor confidence scalar. No calibration parameter or threshold is fitted. These are row-averaged, task-specific diagnostics on inspected synthetic graphs, not population calibration guarantees.

All 792 actions per method are eligible. Jev retains 774 vectors after eighteen sum-invalid degree vectors are omitted; both choice/argmax-disagreeing vectors are retained with the actual supplied choices. All 792 token-score vectors are valid, giving 774 common-valid rows across tasks. Despite Jev’s above-chance action accuracy on distance (48/80) and articulation (47/80), their Brier losses, 0.509155 and 0.508225, exceed the uniform reference of 0.5; their logarithmic losses also exceed  $\log 2$ . Thus accuracy above chance and numeric probability output do not establish useful uncertainty estimates. In contrast, Jev’s adjacency Brier is 0.000120. The task-specific pattern, not a pooled calibration ranking, is the intended diagnostic.

This audit also exposed a bookkeeping bug after inference: canonical JavaScript Object Notation (JSON) key sorting changed the order of floating-point additions. Ten native sum metadata values differed on replay by at most  $2.22 \times 10^{-16}$ , although no vector’s eligibility changed. The original producer failed to seal its native lane; all 792 recorded responses were retained without additional calls. Corrected offline analysis is separately versioned and preserves the original failure, vectors and action decisions. Its derived-sum roundoff check is distinct from both the unchanged  $10^{-6}$  vector gate and the failed 0.01 CPU/GPU reproduction criterion.

**A prospective TSP wording confirmation.** Six late Jev TSP control failures all selected the city closest to start vertex 0, although the actual current endpoint differed. Independent enumeration confirmed the labels. This motivated a bundled state-explicit wording treatment, not a claim to

Table 29: Prospective probability audit: summed multiclass Brier, lower is better. “Valid J/T” gives each method’s own vector denominator; loss columns use exactly the common-valid rows. Uniform is evaluated on the same rows with their actual option cardinalities. No cross-task aggregate is used.

| Task         | Valid J/T | Common | Jev-1.13.0<br>(Native) | Qwen3.5-4B<br>(Candidate scoring) | Uniform |
|--------------|-----------|--------|------------------------|-----------------------------------|---------|
| Adjacency    | 80/80     | 80     | 0.000                  | 0.199                             | 0.500   |
| Degree       | 382/400   | 382    | 0.458                  | 0.841                             | 0.900   |
| Cycle        | 72/72     | 72     | 0.420                  | 0.720                             | 0.500   |
| Connectivity | 80/80     | 80     | 0.152                  | 0.682                             | 0.500   |
| Distance     | 80/80     | 80     | 0.509                  | 0.550                             | 0.500   |
| Articulation | 80/80     | 80     | 0.508                  | 0.534                             | 0.500   |

identify an internal mechanism.

Forty newly generated weighted metrics were independently excluded from eighty earlier weighted TSP instances within a refreshed 1,840-record pool. Under the original strict protocol, Jev completes 36/40 original-wording and 33/40 explicit-wording episodes. On the 29 commonly feasible graphs, relative gap improves by 0.3761 (pointwise 95% interval [0.2337, 0.5462]); this conditional comparison does not establish improved completion.

A separately registered action-observation run on those same forty metrics completes all eighty episodes. Mean relative gap falls from 0.5152 to 0.1603; the paired reduction is 0.3550 [0.2463, 0.4729]. Nine strict-native observations remain recorded. This second protocol removes feasibility selection within that comparison, but is a repeated use of the metrics, not a second untouched confirmation set. No prior failed response was retrospectively rescued.

**Stronger reasoning models change the picture.** Completed earlier *seven-task, legacy-protocol development* runs provide an important counterpoint. GPT-5.4 solves all 200 exact decisions and all 62 feasibly completed construction episodes optimally; two timeouts leave two other constructions incomplete (559 actual calls). GPT-6-Astra solves all 200 exact decisions and 63/64 constructions optimally with no failures (576 calls); its single suboptimal cut has value 20 versus optimum 21. These are not missing columns to paste into the common-interface challenge: the tasks, bank and protocol differ. They nevertheless preclude interpreting weak bounded Qwen results as evidence of universal LLM incapacity.

The subsequently completed *ten-task legacy-interface GPT-6-Astra challenge* uses the same 960 graph instances and references as the inspected canonical challenge. Its frozen runner replays all 1,920 calls with no failures. All 800 exact decisions are correct, and 159/160 constructions are optimal (Table 30). The sole miss is an eight-vertex community partition with  $Q = 0.0244444$  versus  $Q^* = 0.08$ ; independent NetworkX evaluation confirms all forty community objectives, and enumeration of all 128 anchored assignments confirms this optimum.

This result is intentionally separate from the synthetic matrix: the legacy instructions and option descriptions differ, with zero of 960 full initial requests identical to their canonical counterparts. GPT-6-Astra uses default reasoning and a 4,096-token ceiling rather than the bounded local decoding settings. The result demonstrates that these graph instances are tractable for this stronger configuration; it does not isolate model architecture, protocol effects or equal-compute superiority.

Table 30: Completed GPT-6-Astra legacy-interface challenge on the same graphs as the inspected synthetic challenge. The first six rows report correct answers; the final four report optimum hits. Mean gaps are task-specific and are not pooled. This is a separate protocol/default-reasoning reference, not an additional column of the canonical four-method matrix.

| Task             | Correct / optimal | Feasible | Mean gap |
|------------------|-------------------|----------|----------|
| Adjacency        | 80/80             | 80/80    | –        |
| Degree           | 400/400           | 400/400  | –        |
| Cycle            | 80/80             | 80/80    | –        |
| Connectivity     | 80/80             | 80/80    | –        |
| Distance         | 80/80             | 80/80    | –        |
| Articulation     | 80/80             | 80/80    | –        |
| MaxCut           | 40/40             | 40/40    | 0.000000 |
| TSP              | 40/40             | 40/40    | 0.000000 |
| Deterministic LT | 40/40             | 40/40    | 0.000000 |
| Modularity       | 39/40             | 40/40    | 0.001389 |

### Q.1 Completed canonical GPT-5.4 anchor

This separately registered tool-free, default-reasoning GPT-5.4 run uses the identical canonical structural-challenge requests and own histories. It made 1,840 actual calls, with 9 failed attempts. Failures remain in scheduled denominators; no retry or repair was performed. Observed median/p95

Table 31: Canonical exact macro-size accuracy. Intervals are paired, size-stratified pointwise GPT-minus-Jev differences, not family-wise tests.

| Exact task   | GPT-5.4<br>(Default reasoning) (%) | Jev-1.13.0<br>(Native) (%) | 95% interval (pp) |
|--------------|------------------------------------|----------------------------|-------------------|
| Adjacency    | 100.00                             | 100.00                     | [0.00, 0.00]      |
| Degree       | 100.00                             | 61.43                      | [33.75, 43.30]    |
| Cycle        | 100.00                             | 66.25                      | [27.50, 40.00]    |
| Connectivity | 100.00                             | 85.00                      | [8.75, 22.50]     |
| Distance     | 100.00                             | 58.75                      | [35.00, 46.25]    |
| Articulation | 100.00                             | 65.00                      | [26.25, 43.75]    |

Table 32: Canonical construction results. Gaps condition on GPT feasible completion and have different units across rows; they are not a pooled score.

| Task             | GPT-5.4<br>feasible | GPT-5.4<br>optima | Jev-1.13.0<br>optima | GPT-5.4<br>gap |
|------------------|---------------------|-------------------|----------------------|----------------|
| MaxCut           | 36/40               | 36/40             | 0/40                 | 0.000          |
| TSP              | 39/40               | 39/40             | 3/40                 | 0.000          |
| Deterministic LT | 40/40               | 40/40             | 23/40                | 0.000          |
| Modularity       | 36/40               | 36/40             | 7/40                 | 0.000          |

GPT request latency was 7220.20/96561.70 ms. Reasoning budget and hosting differ from the small no-thinking local baselines. These are inspected-bank common-protocol anchors, not the prospective new-seed confirmation or a matched-compute architectural comparison. Request identity here concerns all 960 initial model-visible requests. Later construction requests follow each method’s own choices and need not match; neither model receives the other’s trajectory or oracle actions.

All nine failed attempts are timeouts in construction episodes: four MaxCut, one TSP and four modularity. Across 960 scheduled episodes, all 800 exact answers are correct and all 151 feasible constructions attain exact optima. An additional independent objective check reproduces every feasible construction using NetworkX or exact-rational diffusion. This strengthens the interpretation that the bounded models’ failures are not unavoidable properties of the graph instances or the canonical action interface. It does not establish equal-compute superiority, out-of-distribution generalization, or unconditional construction perfection: the nine incomplete episodes remain in every all-scheduled denominator.

**Exploratory larger-local-model check.** After inspecting the synthetic results, we registered a small, score-blind selection from the confirmation bank: four cases per exact task and two per construction at each of  $n = 8, 12$ , retaining binary pairs and degree strata. The resulting 64-episode, 160-decision panel compares a fresh Qwen3.5-27B BF16 run with reused Jev and bounded-Qwen reference traces on exactly the same instances. The 27B configuration keeps the 64-token, no-thinking, legal-choice grammar contract and completes every episode without failure. This is an inspected-bank exploratory extension, not an independent replication.

On this panel Jev exceeds bounded 27B on degree (6/8 versus 4/8), connectivity (7/8 versus 6/8), and cycle detection (5/8 versus 4/8), while 27B exceeds Jev on distance (5/8 versus 4/8). Adjacency and articulation tie (Appendix W). Jev has better mean cut, TSP and community objectives; LT ties, including 2/4 optimum hits for both. Neither reaches any cut, TSP or community optimum on these four-case tasks. These observations concern the specified no-thinking grammar configuration only. They do not establish parameter equivalence or bound 27B’s unrestricted capacity; the completed thinking comparison below substantially changes the exact-task interpretation. Historical reference latency is not a controlled timing comparison, and these small denominators support no significance claim.

**Thinking improves exact answers but exposes deadline failures.** The completed 27B and 9B thinking lanes use the same selected instances and canonical initial requests, but enable thinking, remove grammar constraints, and raise the output-token cap from 64 to 8,192 jointly. The per-request deadline is 180 seconds. Their exact counts are 47/48 and 40/48, versus 33/48 for reused Jev and 30/48 for 27B grammar. Every returned exact answer is correct: 47/47 and 40/40, respectively. The remaining exact cases are timeouts, not observed wrong predictions; 27B’s one timeout is a distance query. All 48 scheduled exact cases remain in the reported accuracy denominators.

Both thinking lanes time out within every one of their sixteen construction episodes, leaving zero feasible constructions. Across the whole panel, 27B records 17 timed-out calls among 66 attempts; 9B records 24 among 64. Conditional construction gaps are therefore undefined, not zero, and provide no quality ranking between these configurations and methods that finish. The substantially stronger exact results overturn an interpretation of the bounded comparison as Jev–27B capability equivalence. This is a joint inference-configuration effect, not an isolated causal effect of reasoning.

A separate timeout-only 9B control raises the deadline to 600 seconds while retaining the 8,192-token cap. All sixteen construction attempts then terminate at that token cap, still yielding no feasible trajectories (Table 40). Increasing wall time alone therefore does not resolve these completion failures.

**Two additional trained decision readouts.** We also evaluate the released Decider-2B v11 and Kev-4B trained inference paths [9, 10, 15, 16], not ordinary next-token substitutes. Both complete

all 64 episodes in 160 calls, without execution failures. Decider obtains 24/48 exact answers and Kev 23/48, versus 33/48 for Jev. Neither answers a degree query correctly, despite legal, varied predictions and the full eight- or twelve-class vocabulary. Decider reaches no construction optimum; Kev reaches one LT optimum, compared with Jev’s two LT optima. This is not an all-task Jev win: Kev’s mean modularity gap is 0.113657, better than Jev’s 0.235880. Appendix W reports taskwise coverage, optimum hits and conditional gaps separately, along with dtype, source-version and environment limitations. Native probability conformance is not evidence of calibration on these graph tasks.

**An encoder-based trained decision model.** A separate Laya typed-decisions pilot evaluates a roughly 421-million-parameter ModernBERT encoder with its genuine trained choice head [4], using native CPU FP32 inference on the same 64-case panel. Its checkpoint contract permits 1,024 total tokens and a 256-token head. Lossless tokenization checks reject two  $n = 12$  LT requests before inference; no graph, question or option is truncated. The remaining 62 episodes require 156 successful forwards, with no later context overflow. Laya obtains 22/48 exact answers versus 33/48 for reused Jev, and no optimal construction among fourteen initially supported cases versus two for Jev. All forty binary answers are **yes**, yielding the constant-yes baseline of 20/40; degree contributes 2/8. Native logits, probability-to-ID mapping and canonical action replay are checked rather than assuming these constant predictions are a parser failure. A separate twenty-call option-order diagnostic preserves **yes** on all ten paired states after reversal, arguing against a fixed-first-option explanation on this selected subset, not establishing a claim about every Laya prompt or task. Appendix X separates input coverage from solution quality and discloses an offline postprocessing recovery.

## R Synthetic exact contracts and uniform-policy references

This appendix concerns the small synthetic tasks. Public-instance scoring uses the distinct contracts in Appendix F. Node relabeling does not change truth, but rendering may affect model outputs. Cycle twins match degree sequences, not component count. With  $c(G)$  denoting the number of connected components,  $m > n - c(G)$  is an exact cycle test for the stated undirected contract. Distance pairs share their threshold and finite distances  $h, h + 1$ . Articulation compares total component counts before and after deletion; isolated vertices are not articulation points under this definition.

For LT, normalized incoming weights and a fixed one-half threshold are applied synchronously to the two initially chosen seeds. Seeds are not activated one at a time with diffusion between selections. For TSP, prefix cost, the current endpoint, all remaining cities, and final return to 0 enter continuation values. For partitions, vertex 0 fixes complement symmetry without excluding any attainable objective; empty sides remain legal. Modularity search is over at most two groups, including one group. Histories must contain legal, exact-string action IDs, not a repaired subset of malformed model output.

Synthetic reference values are obtained by exhaustive partitions/seed pairs or exact TSP dynamic programming and checked independently. Uniform cut expectation is  $m/2$ ; uniform TSP expectation is  $2 \sum_{\{u,v\}} d_{uv} / (n - 1)$ . Uniform LT averages all distinct unordered seed pairs. For uniform binary modularity with one fixed vertex, expected modularity is  $-\sum_v d_v^2 / (8m^2)$ . These are expectations of random legal policies, not attainable objective guarantees for one particular rollout. For a signed weighted cut, the corresponding uniform expectation would be  $\frac{1}{2} \sum_{\{u,v\} \in E} w_{uv}$ , not  $m/2$ . The public panel instead reports one sampled random construction per instance and condition.

**Why the LT guarantee cannot be imported.** Consider vertices  $a, b, d, c$  with edges  $a \rightarrow c$ ,  $b \rightarrow c$ ,  $d \rightarrow c$ . Each edge has weight  $1/3$  and  $c$  requires at least two active parents. Writing  $\sigma(S)$  for final spread including seeds,  $\sigma(\emptyset) = 0$ ,  $\sigma(\{a\}) = \sigma(\{b\}) = 1$ , and  $\sigma(\{a, b\}) = 3$ . The marginal gain of adding  $b$  is 1 at the empty set but 2 at  $\{a\}$ , violating diminishing returns. This four-vertex example uses the benchmark’s exact deterministic diffusion semantics and illustrates why the randomized-threshold theorem does not apply.

## S Paired uncertainty for the inspected matrix

| Task             | $\Delta$ (pp) | Pointwise 95% interval | Clusters |
|------------------|---------------|------------------------|----------|
| Adjacency        | 40.00         | [33.75, 46.25]         | 40       |
| Degree           | 43.11         | [37.46, 48.88]         | 400      |
| Cycle            | 11.25         | [2.50, 18.75]          | 40       |
| Connectivity     | 35.00         | [27.50, 41.25]         | 40       |
| Distance         | 8.75          | [3.75, 15.00]          | 40       |
| Articulation     | 15.00         | [6.25, 23.75]          | 40       |
| MaxCut           | 0.00          | [0.00, 0.00]           | 40       |
| TSP              | 7.50          | [0.00, 15.00]          | 40       |
| Deterministic LT | 25.00         | [5.00, 42.50]          | 40       |
| Modularity       | 17.50         | [7.50, 30.00]          | 40       |

Differences are Jev action minus 4B grammar in percentage points: exact accuracy for the first six rows and optimum-hit rate for the last four. Resampling uses paired identity/construction clusters within each size. Forty clusters means eight per size for these paired binary and optimization tasks. Degree has 400 clusters. The MaxCut zero-width interval reflects the observed all-zero optimum-hit difference, not established equivalence of the underlying models or equal gap quality.

## T Paired uncertainty for prospective confirmation

| Task             | $\Delta$ (pp) | Pointwise 95% interval | Clusters |
|------------------|---------------|------------------------|----------|
| Adjacency        | 36.25         | [30.00, 42.50]         | 40       |
| Degree           | 46.77         | [41.07, 52.73]         | 400      |
| Cycle            | 20.00         | [10.00, 30.00]         | 36       |
| Connectivity     | 37.50         | [31.25, 43.75]         | 40       |
| Distance         | 8.75          | [3.75, 15.00]          | 40       |
| Articulation     | 8.75          | [0.00, 17.50]          | 40       |
| MaxCut           | 0.00          | [0.00, 0.00]           | 40       |
| TSP              | 5.00          | [0.00, 10.00]          | 40       |
| Deterministic LT | 37.50         | [22.50, 52.50]         | 40       |
| Modularity       | 5.00          | [0.00, 12.50]          | 40       |

Conventions are the same as the preceding appendix, but these are new graphs under the pre-inference registration. Cycle has 36 paired clusters, with four at size 8 and eight at each other size. No missing pair is assigned a synthetic model prediction. The construction tasks retain forty clusters each.

## U Presentation-control detail

Tables below report per-arm accuracy and semantic-choice changes for the same selected queries; equal accuracy need not imply equal decisions.

Table 33: Correct cases / scheduled cases in each arm. Counts are not independent across arms or paired queries. An unchanged aggregate accuracy does not imply unchanged semantic choices.

| Task         | Method                         | Original | Repeat | Relabel | Edges | Options |
|--------------|--------------------------------|----------|--------|---------|-------|---------|
| Adjacency    | Jev-1.13.0 (Native)            | 20/20    | 20/20  | 20/20   | 20/20 | 20/20   |
| Adjacency    | Qwen3.5-4B (Grammar)           | 14/20    | 14/20  | 12/20   | 13/20 | 19/20   |
| Adjacency    | Qwen3.5-9B (Grammar)           | 12/20    | 12/20  | 14/20   | 13/20 | 14/20   |
| Adjacency    | Qwen3.5-4B (Candidate scoring) | 17/20    | 17/20  | 15/20   | 19/20 | 14/20   |
| Degree       | Jev-1.13.0 (Native)            | 15/20    | 14/20  | 11/20   | 12/20 | 13/20   |
| Degree       | Qwen3.5-4B (Grammar)           | 3/20     | 3/20   | 2/20    | 2/20  | 6/20    |
| Degree       | Qwen3.5-9B (Grammar)           | 9/20     | 9/20   | 7/20    | 10/20 | 9/20    |
| Degree       | Qwen3.5-4B (Candidate scoring) | 8/20     | 8/20   | 6/20    | 7/20  | 10/20   |
| Cycle        | Jev-1.13.0 (Native)            | 15/20    | 15/20  | 13/20   | 13/20 | 14/20   |
| Cycle        | Qwen3.5-4B (Grammar)           | 10/20    | 10/20  | 10/20   | 10/20 | 10/20   |
| Cycle        | Qwen3.5-9B (Grammar)           | 11/20    | 11/20  | 10/20   | 10/20 | 10/20   |
| Cycle        | Qwen3.5-4B (Candidate scoring) | 10/20    | 10/20  | 10/20   | 10/20 | 11/20   |
| Connectivity | Jev-1.13.0 (Native)            | 18/20    | 18/20  | 15/20   | 15/20 | 19/20   |
| Connectivity | Qwen3.5-4B (Grammar)           | 10/20    | 10/20  | 10/20   | 10/20 | 10/20   |
| Connectivity | Qwen3.5-9B (Grammar)           | 10/20    | 10/20  | 10/20   | 10/20 | 10/20   |
| Connectivity | Qwen3.5-4B (Candidate scoring) | 10/20    | 10/20  | 10/20   | 10/20 | 9/20    |
| Distance     | Jev-1.13.0 (Native)            | 11/20    | 11/20  | 11/20   | 12/20 | 11/20   |
| Distance     | Qwen3.5-4B (Grammar)           | 10/20    | 10/20  | 10/20   | 10/20 | 10/20   |
| Distance     | Qwen3.5-9B (Grammar)           | 10/20    | 10/20  | 10/20   | 9/20  | 10/20   |
| Distance     | Qwen3.5-4B (Candidate scoring) | 11/20    | 11/20  | 10/20   | 10/20 | 10/20   |
| Articulation | Jev-1.13.0 (Native)            | 11/20    | 11/20  | 11/20   | 11/20 | 10/20   |
| Articulation | Qwen3.5-4B (Grammar)           | 10/20    | 10/20  | 10/20   | 10/20 | 10/20   |
| Articulation | Qwen3.5-9B (Grammar)           | 10/20    | 10/20  | 10/20   | 10/20 | 10/20   |
| Articulation | Qwen3.5-4B (Candidate scoring) | 10/20    | 10/20  | 10/20   | 10/20 | 10/20   |

Table 34: Choice flips relative to the original arm / both-valid pairs. There are no invalid actions in this completed panel. Edge counts include the two retained unchanged inputs; their separate strata are in the artifact.

| Task         | Method                         | Repeat | Relabel | Edges | Options |
|--------------|--------------------------------|--------|---------|-------|---------|
| Adjacency    | Jev-1.13.0 (Native)            | 0/20   | 0/20    | 0/20  | 0/20    |
| Adjacency    | Qwen3.5-4B (Grammar)           | 0/20   | 2/20    | 3/20  | 7/20    |
| Adjacency    | Qwen3.5-9B (Grammar)           | 0/20   | 4/20    | 3/20  | 2/20    |
| Adjacency    | Qwen3.5-4B (Candidate scoring) | 0/20   | 4/20    | 2/20  | 9/20    |
| Degree       | Jev-1.13.0 (Native)            | 2/20   | 6/20    | 4/20  | 2/20    |
| Degree       | Qwen3.5-4B (Grammar)           | 0/20   | 12/20   | 11/20 | 12/20   |
| Degree       | Qwen3.5-9B (Grammar)           | 0/20   | 13/20   | 6/20  | 9/20    |
| Degree       | Qwen3.5-4B (Candidate scoring) | 0/20   | 13/20   | 6/20  | 14/20   |
| Cycle        | Jev-1.13.0 (Native)            | 0/20   | 4/20    | 4/20  | 3/20    |
| Cycle        | Qwen3.5-4B (Grammar)           | 0/20   | 6/20    | 8/20  | 8/20    |
| Cycle        | Qwen3.5-9B (Grammar)           | 0/20   | 1/20    | 1/20  | 1/20    |
| Cycle        | Qwen3.5-4B (Candidate scoring) | 0/20   | 0/20    | 0/20  | 1/20    |
| Connectivity | Jev-1.13.0 (Native)            | 0/20   | 5/20    | 3/20  | 1/20    |
| Connectivity | Qwen3.5-4B (Grammar)           | 0/20   | 0/20    | 0/20  | 0/20    |
| Connectivity | Qwen3.5-9B (Grammar)           | 0/20   | 0/20    | 0/20  | 0/20    |
| Connectivity | Qwen3.5-4B (Candidate scoring) | 0/20   | 0/20    | 0/20  | 3/20    |
| Distance     | Jev-1.13.0 (Native)            | 0/20   | 0/20    | 1/20  | 2/20    |
| Distance     | Qwen3.5-4B (Grammar)           | 0/20   | 0/20    | 0/20  | 0/20    |
| Distance     | Qwen3.5-9B (Grammar)           | 0/20   | 6/20    | 7/20  | 6/20    |
| Distance     | Qwen3.5-4B (Candidate scoring) | 0/20   | 5/20    | 7/20  | 13/20   |
| Articulation | Jev-1.13.0 (Native)            | 2/20   | 2/20    | 0/20  | 1/20    |
| Articulation | Qwen3.5-4B (Grammar)           | 0/20   | 0/20    | 0/20  | 18/20   |
| Articulation | Qwen3.5-9B (Grammar)           | 0/20   | 0/20    | 0/20  | 20/20   |
| Articulation | Qwen3.5-4B (Candidate scoring) | 0/20   | 0/20    | 0/20  | 20/20   |

## V Probability-loss detail

Table 35: Logarithmic loss on the common-valid rows from Table 29, in natural-log units. A positive-infinity entry means at least one true label received exactly zero probability, not a missing value. Finite-only averaging is not substituted for an infinite loss.

| Task         | Common | Jev-1.13.0<br>(Native) | Qwen3.5-4B<br>(Candidate scoring) | Uniform |
|--------------|--------|------------------------|-----------------------------------|---------|
| Adjacency    | 80     | 0.002                  | 0.331                             | 0.693   |
| Degree       | 382    | 0.935                  | 2.036                             | 2.312   |
| Cycle        | 72     | 0.616                  | 1.045                             | 0.693   |
| Connectivity | 80     | 0.229                  | 1.009                             | 0.693   |
| Distance     | 80     | 0.735                  | 0.747                             | 0.693   |
| Articulation | 80     | 0.713                  | 0.729                             | 0.693   |

The artifact includes all five reliability-bin counts and conditional accuracies for each task and method, along with all-scheduled action accuracy, own-valid and common-valid denominators. Vector eligibility is not assumed independent of task difficulty. The 4B vectors are conditional option-logit scores, not calibrated posterior probabilities by definition.

## W Expanded-model exploratory comparison

The selected graphs are reused from the confirmation bank; opaque-ID selection within prespecified pairs/strata does not make this a new held-out evaluation. The 27B configurations use tensor

parallelism across two RTX A6000s. Initial model-visible requests are identical across compared configurations, but provider interfaces differ and subsequent requests follow each method’s own history. The grammar and trained-native lanes complete all selected episodes; the thinking lanes have the explicit failures below. Each original historical lane is fully replayed before subsetting. Terminal candidate requests, actions, all 64 score rows and summaries are independently replayed and hash-attested. NetworkX/Fraction checks reproduce every feasible construction objective. Before manuscript export, offline checks verified source hashes, recomputed the denominators and metrics from comma-separated values (CSV) summaries, and required identical reused reference trajectories across all exports. No failed episode is removed.

Table 36: Exact correct/scheduled answers on the same selected panel. Each task has four cases at each of  $n = 8, 12$ . Grammar uses a 64-token no-thinking setting; Thinking uses unconstrained generation with an 8,192-token cap and 180-second deadline. Jev is a reused reference, not fresh inference. Thinking timeouts remain in the denominators; these are descriptive counts, not a population ranking.

| Task         | Jev-1.13.0<br>(Native) | Qwen3.5-27B<br>(Grammar) | Qwen3.5-27B<br>(Thinking) | Qwen3.5-9B<br>(Thinking) | Decider-2B<br>(Native) | Kev-4B<br>(Native) |
|--------------|------------------------|--------------------------|---------------------------|--------------------------|------------------------|--------------------|
| Adjacency    | 8/8                    | 8/8                      | 8/8                       | 8/8                      | 7/8                    | 7/8                |
| Degree       | 6/8                    | 4/8                      | 8/8                       | 7/8                      | 0/8                    | 0/8                |
| Cycle        | 5/8                    | 4/8                      | 8/8                       | 8/8                      | 4/8                    | 4/8                |
| Connectivity | 7/8                    | 6/8                      | 8/8                       | 7/8                      | 4/8                    | 4/8                |
| Distance     | 4/8                    | 5/8                      | 7/8                       | 7/8                      | 5/8                    | 5/8                |
| Articulation | 3/8                    | 3/8                      | 8/8                       | 3/8                      | 4/8                    | 3/8                |
| Exact total  | 33/48                  | 30/48                    | 47/48                     | 40/48                    | 24/48                  | 23/48              |

Table 37: Completion coverage, separate from correctness. “Failed” counts episodes with an execution failure, all timeouts in the two thinking lanes. “Exact returned” counts feasible exact episodes, not correct answers. Every configuration retains all 64 scheduled episodes. All grammar/native configurations use 160 calls; Qwen3.5-27B and Qwen3.5-9B with thinking use 66 and 64, respectively, because timeouts terminate individual trajectories without retries.

| Configuration          | Scheduled | Failed | Exact returned | Constructed |
|------------------------|-----------|--------|----------------|-------------|
| Jev-1.13.0 (Native)    | 64        | 0      | 48/48          | 16/16       |
| Qwen3.5-27B (Grammar)  | 64        | 0      | 48/48          | 16/16       |
| Qwen3.5-27B (Thinking) | 64        | 17     | 47/48          | 0/16        |
| Qwen3.5-9B (Thinking)  | 64        | 24     | 40/48          | 0/16        |
| Decider-2B (Native)    | 64        | 0      | 48/48          | 16/16       |
| Kev-4B (Native)        | 64        | 0      | 48/48          | 16/16       |

**Deadline and output-budget interpretation.** The 8,192-token cap is not a guarantee of that much decoding before the 180-second wall deadline. For example, 21.3 generated tokens/second permits only about 3,834 tokens in 180 seconds even before other overhead; this illustrative arithmetic is not a controlled speed comparison. A separately registered posthoc Qwen3.5-9B control evaluates all sixteen constructions; only the deadline changes to 600 seconds, with the 8,192-token cap unchanged (at most 112 calls). All sixteen first-step attempts end with `finish_length`, each recording exactly 8,192 output tokens and approximately 239 seconds of client-observed latency, below the relaxed deadline. No response finishes normally, so no action

Table 38: Four constructions per task (two at each size). Feasibility and all-scheduled optimum-hit rates are distinct outcomes. The thinking lanes’ 0/4 optimum counts arise from zero completed constructions, not four observed suboptimal solutions. Labels follow Table 36.

| Feasible / scheduled |                        |                          |                           |                          |                        |                    |
|----------------------|------------------------|--------------------------|---------------------------|--------------------------|------------------------|--------------------|
| Task                 | Jev-1.13.0<br>(Native) | Qwen3.5-27B<br>(Grammar) | Qwen3.5-27B<br>(Thinking) | Qwen3.5-9B<br>(Thinking) | Decider-2B<br>(Native) | Kev-4B<br>(Native) |
| MaxCut               | 4/4                    | 4/4                      | 0/4                       | 0/4                      | 4/4                    | 4/4                |
| TSP                  | 4/4                    | 4/4                      | 0/4                       | 0/4                      | 4/4                    | 4/4                |
| Deterministic LT     | 4/4                    | 4/4                      | 0/4                       | 0/4                      | 4/4                    | 4/4                |
| Modularity           | 4/4                    | 4/4                      | 0/4                       | 0/4                      | 4/4                    | 4/4                |

| Optimum hits / scheduled |                        |                          |                           |                          |                        |                    |
|--------------------------|------------------------|--------------------------|---------------------------|--------------------------|------------------------|--------------------|
| Task                     | Jev-1.13.0<br>(Native) | Qwen3.5-27B<br>(Grammar) | Qwen3.5-27B<br>(Thinking) | Qwen3.5-9B<br>(Thinking) | Decider-2B<br>(Native) | Kev-4B<br>(Native) |
| MaxCut                   | 0/4                    | 0/4                      | 0/4                       | 0/4                      | 0/4                    | 0/4                |
| TSP                      | 0/4                    | 0/4                      | 0/4                       | 0/4                      | 0/4                    | 0/4                |
| Deterministic LT         | 2/4                    | 2/4                      | 0/4                       | 0/4                      | 0/4                    | 1/4                |
| Modularity               | 0/4                    | 0/4                      | 0/4                       | 0/4                      | 0/4                    | 0/4                |

Table 39: Mean additive objective gap conditional on feasibility, lower being better. Each numeric cell has four feasible episodes; each dash denotes zero feasible episodes and an undefined mean, not zero gap. Units are cut edges, tour cost, active vertices and modularity respectively; they are never pooled. Kev improves the modularity gap over Jev despite lower exact-task accuracy.

| Task             | Jev-1.13.0<br>(Native) | Qwen3.5-27B<br>(Grammar) | Qwen3.5-27B<br>(Thinking) | Qwen3.5-9B<br>(Thinking) | Decider-2B<br>(Native) | Kev-4B<br>(Native) |
|------------------|------------------------|--------------------------|---------------------------|--------------------------|------------------------|--------------------|
| MaxCut           | 8.25                   | 9.00                     | —                         | —                        | 12.25                  | 12.25              |
| TSP              | 60.50                  | 146.00                   | —                         | —                        | 387.50                 | 360.00             |
| Deterministic LT | 1.00                   | 1.00                     | —                         | —                        | 4.00                   | 1.50               |
| Modularity       | 0.235880               | 0.305421                 | —                         | —                        | 0.305421               | 0.113657           |

advances a trajectory. We do not salvage a choice from truncated final content, retry, or raise the token cap. All sixteen constructions remain infeasible. Thus, under this registered contract, relaxing the wall deadline exposes a token-limit failure rather than a feasible solution-quality estimate. This result is specific to the tested prompt, budget and inference setting; it does not establish inability to construct solutions under other settings.

**Pinned trained methods and compatibility.** Decider uses its genuine one-pass restricted-letter readout, BF16 weights and eager inference. The v11 checkpoint requires source version 1.4.0 (commit [15ab28e8](#)); its plain-choice temperature is 1.164, not the global fallback 1.145. Kev uses the pinned upstream pointer head and FP32 backbone merged with low-rank adaptation (LoRA) (source commit [569ea449](#)), with temperature 2.406050072164233 and an independently pinned Qwen3.5-4B-Base. Model-card and code revisions are cited separately [9, 10, 15, 16]. All 34 registered checkpoint/tokenizer/configuration files are hash-checked; temperatures are independently derived from the Decider configuration and restricted CPU loading of the Kev head metadata, rather than trusted solely from GPU runtime claims. Complete requests retain all original semantic option IDs and descriptions, with an 8,192-token lossless context gate.

Table 40: Completed timeout-only Qwen3.5-9B control on the same sixteen constructions. All four cases per task remain in denominators. The 180-second arm has wall-timeout failures; the 600-second arm has token-limit failures. Conditional objective gaps are undefined in both arms, not zero.

| Task             | 180s feasible | 600s feasible | 600s length stops | 600s gap |
|------------------|---------------|---------------|-------------------|----------|
| MaxCut           | 0/4           | 0/4           | 4                 | –        |
| TSP              | 0/4           | 0/4           | 4                 | –        |
| Deterministic LT | 0/4           | 0/4           | 4                 | –        |
| Modularity       | 0/4           | 0/4           | 4                 | –        |

The compatible shared runtime uses Python 3.12, Torch 2.13.0, Transformers 5.17.0 and NumPy 2.3.5, with isolated Accelerate 1.15.0 and Parameter-Efficient Fine-Tuning (PEFT) 0.21.0 packages. These are not the full declared upstream environments: Kev specifies Torch below 2.9 and NumPy at least 2.5.3, while Decider specifies NumPy below 2. Unavailable fused kernels use native reference paths. Dtype and deployment are not matched across methods; no latency ranking or full-environment reproduction is claimed. Released temperature fitting is not a calibration guarantee on this selected graph panel.

**A limited CPU/GPU numeric check.** A separate posthoc control, registered before its fresh CPU forwards, repeats the first  $n = 8$  adjacency and first  $n = 12$  degree request for each trained model. All four forwards pass the fixed gate of maximum absolute probability difference at most 0.01 plus native-choice agreement. In adjacency/degree order, the differences are approximately  $1.49 \times 10^{-8}$  and 0.002753 for Decider, and  $1.19 \times 10^{-7}$  and  $2.91 \times 10^{-7}$  for Kev. Option keys, argmaxes and input-token counts also agree. The original serialized model-visible inputs are reconstructed without changing values or options; CPU workers use four threads with the GPU runtime uninitialized, and no new GPU inference is performed. This is compatibility on only two requests per model, not exact equality, broad device parity, a quality/calibration guarantee or full-environment reproduction. Agreement also reproduces the incorrect degree choices; it does not make them correct. The earlier SemIf candidate-token failed numerical gate is a distinct experiment and remains unresolved.

**Validated failures versus mathematical errors.** Both trained methods’ eight degree predictions vary and remain within the full legal eight-/twelve-class sets. Recorded option-to-probability mappings and own-history replay agree, and neither prediction vector is explained by a global +1 or −1 answer offset. On MaxCut, both always assign every new vertex to B; vertex 0 remains fixed in A. Decider assigns all community vertices to A, yielding verified zero modularity. These feasible objective and exact-answer errors are not parser failures. Conversely, thinking construction timeouts provide no realized completed objective. The four-case construction denominators remain too small for formal significance or universal capability claims.

## X Encoder-based typed-decision pilot

The checkpoint and inference code are pinned separately. The 16-bit floating-point (F16) checkpoint weights are loaded by the upstream CPU path into FP32 without automatic mixed precision (AMP); this is not a GPU timing or equal-precision comparison. Four CPU threads are used, with a 30-second forward deadline and a separate 30-minute worker budget. Initial coverage is

62/64. Both unsupported LT cases exceed the native option-head budget; one also exceeds the total context limit. All 64 remain in scheduled denominators, giving 0/16 Laya and 2/16 Jev construction optimum hits overall. Unsupported inputs are not wrong predictions. Table quality comparisons use the fourteen initially supported constructions, not a retrospectively chosen easy subset.

Table 41: Laya CPU exact-task results on the exploratory panel. All eight instances per task fit losslessly. Historical Jev is independently replayed; these small differences are not a population ranking.

| Task         | Laya<br>(Native) CPU | Jev-1.13.0<br>Reused (Native) |
|--------------|----------------------|-------------------------------|
| Adjacency    | 4/8                  | 8/8                           |
| Degree       | 2/8                  | 6/8                           |
| Cycle        | 4/8                  | 5/8                           |
| Connectivity | 4/8                  | 7/8                           |
| Distance     | 4/8                  | 4/8                           |
| Articulation | 4/8                  | 3/8                           |

Table 42: Initial Laya coverage out of four scheduled constructions per task, followed by optimum hits and mean additive gaps on exactly that supported subset for both methods. LT quality uses only the two  $n = 8$  instances. Lower gaps are better, but their units differ across tasks.

| Task             | Covered | Laya<br>(Native) hits | Jev-1.13.0<br>(Native) hits | Laya<br>(Native) gap | Jev-1.13.0<br>(Native) gap |
|------------------|---------|-----------------------|-----------------------------|----------------------|----------------------------|
| MaxCut           | 4/4     | 0/4                   | 0/4                         | 15.250               | 8.250                      |
| TSP              | 4/4     | 0/4                   | 0/4                         | 368.500              | 60.500                     |
| Deterministic LT | 2/4     | 0/2                   | 2/2                         | 3.500                | 0.000                      |
| Modularity       | 4/4     | 0/4                   | 0/4                         | 0.318                | 0.236                      |

The native API rounds probabilities to four decimals and distinguishes entropy concentration from maximum option probability. Rounded vectors are preserved without renormalization or temperature fitting; neither confidence definition is assumed calibrated. The inference ledger was fully persisted, but an in-memory postprocessor failed on tuple-versus-JSON-list representation in model-visible requests. A separate finalizer replayed the durable JSON ledger without changing inference, the frozen source, registration or original error records. All 64 scores and fourteen feasible construction objectives were independently checked. The exact supervisor wall timer was unavailable after that failure and remains missing rather than reconstructed as a benchmark latency.

**Posthoc constant-answer diagnostic.** After observing the all-yes pattern, we registered a separate control: for each binary task at  $n = 8$ , choose the lexicographically first positive and negative instance, then evaluate each in original and reversed option order. Semantic IDs and descriptions are retained, and labels are not model inputs. All twenty calls complete: original order yields ten yes choices at index zero; reversed order yields ten yes choices at index one. Both orders score 5/10. Semantic choices agree on 10/10 pairs, whereas positional indices agree on 0/10. Original-order repeats reproduce the initial-run logits exactly, and all forty initial binary logit vectors are distinct. Full native mask-to-ID checks and an independent ledger verifier support a semantic-yes preference rather than a fixed-first-position or observed mapping error in these states. Reversal still changes score values, so this is not a general absence-of-position-bias claim. The ten

paired states are adaptively selected diagnostics, not twenty independent examples or a replacement for the initial results.

## Y Reproducibility

**Data and recorded decisions.** Local archives separate model-visible inputs, privileged references and action histories. They support mathematical replay, not independent reproduction of provider behavior or latency.

**Verification and reconstruction.** Offline verification checks input hashes, initial-request equality, action histories, scores and scheduled denominators. IDs, actions, counts and statuses are checked exactly; aggregate floating-point comparisons allow absolute error of  $10^{-12}$ , distinct from the failed 0.01 CPU/GPU probability check. Bundle checksums and execution records are retained in local provenance manifests. The BF16 extension pins Qwen3.8-27B to `1d4bf0f2ff6012fd82039f2fa52739d0dd7c60c0` and Qwen2.5-72B-Instruct to `495f39366efef23836d0cfae4fbe635880d2be31`. The standalone manuscript archive rebuilds the document without inference, but is not a complete experimental replay package.